ARIN7015/MATH6015: Topics in Artificial Intelligence and Machine Learning

Statistical Learning Theory

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Outline

1. Supervised Machine Learning
2. Optimization and Generalization
3. Concentration Inequality
4. Complexity Measure
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5. Applications
   - Applications to Support Vector Machines
   - Applications to Shallow Neural Networks
   - Growth Function and VC Dimension for Binary Classification
   - Covering Number
6. Summary

Supervised Machine Learning

Problem Setup

- Sample space $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$ with a probability distribution $\mathbb{P}$
  - Input space $\mathcal{X}$: images, sounds, videos, text, proteins, web pages, social networks, sensors from industry
  - Output space $\mathcal{Y}$: binary labels $\mathcal{Y} = \{\pm 1\}$, real response $\mathcal{Y} = \mathbb{R}$, multiclass $\mathcal{Y} = \{1, \ldots, k\}$, more generally structured outputs
  - ( ,dog), ( ,`car`), ( ,`airplane`), ...

- Samples: n independent and identical draws of $\mathbb{P}$, $\mathbf{z}_i = (\mathbf{x}_i, y_i)$

$$S = \{\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_n\} \tag{1}$$

- Main goal: find a model by fitting the samples so that it can be used for future prediction
  - Parametric models: linear models, neural networks, polynomials
  - Nonparametric models: decision trees, k-nearest neighbors

Hypothesis Space

A hypothesis space $\mathcal{H}$ is a collection of functions from $\mathcal{X} \mapsto \mathcal{Y}$.

Examples

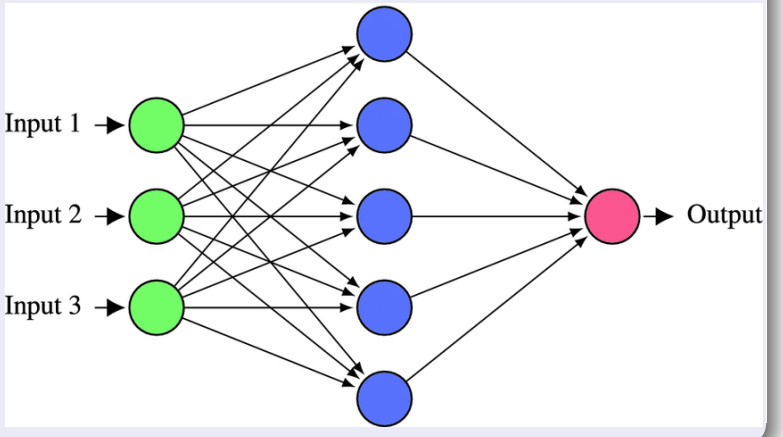
- Linear functions ($\|\cdot\|_2$ is the Euclidean norm)

$$\mathcal{H} = \left\{ \mathbf{x} \mapsto \mathbf{w}^\top \mathbf{x} : \|\mathbf{w}\|_2 \leq 1 \right\}$$

- Shallow neural networks

$$\mathcal{H} = \left\{ \mathbf{x} \mapsto \sum_{j=1}^{m} a_j \sigma\left(\mathbf{w}_j^\top \mathbf{x}\right) : \sum_{j=1}^{m} \|\mathbf{w}_j\|_2^2 \leq 1 \right\},$$

where $\sigma(a) = \max\{a, 0\}$ and $\mathbf{w} = (\mathbf{w}_1, \ldots, \mathbf{w}_m)$.



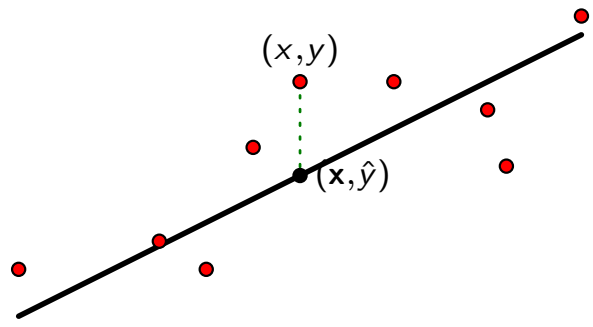
- In this course, we always consider parametric models.
- Each $h \in \mathcal{H}$ is indexed by a parameter $\mathbf{w} \in \mathcal{W}$, where $\mathcal{W}$ is a set of parameters.

Loss Function

- We measure the performance of a model h by a loss function $\ell : \mathbb{R} \times \mathbb{R} \mapsto \mathbb{R}_+$
- It measures the discrepancy between the true output y and the prediction $h(\mathbf{x})$

Regression: for regression we require $\hat{y} = h(\mathbf{x})$ to be close to y. Then a good choice is the distance-based loss

$$\ell(\hat{y}, y) = \frac{1}{2}(\hat{y} - y)^2.$$



Classification: for classification with $\mathcal{Y} = \{1, -1\}$ we often predict based on the sign of $\hat{y} = h(\mathbf{x})$, i.e.,

$$\text{we predict } \begin{cases} 1, & \text{if } \hat{y} \geq 0 \\ -1, & \text{otherwise.} \end{cases}$$

The performance of a model $\mathbf{x}$ on $(\mathbf{x}, y)$ can be measured by the 0-1 loss

$$\ell(y, \hat{y}) = \mathbb{I}[\hat{y} \neq y] = \begin{cases} 1, & \text{if } \hat{y} \neq y \\ 0, & \text{otherwise.} \end{cases}$$

Margin-based Loss

The 0-1 loss is discrete and very difficult to minimize!

$$\hat{y} = \begin{cases} 1, & \text{if } \mathbf{w}^\top \mathbf{x} > 0 \\ -1, & \text{otherwise.} \end{cases} \implies \begin{cases} y = \hat{y} = 1, & \text{if } y = 1, y(\mathbf{w}^\top \mathbf{x}) > 0 \\ y = \hat{y} = -1, & \text{if } y = -1, y(\mathbf{w}^\top \mathbf{x}) \geq 0 \\ y = 1, \hat{y} = -1, & \text{if } y = 1, y(\mathbf{w}^\top \mathbf{x}) \leq 0 \\ y = -1, \hat{y} = 1, & \text{if } y = -1, y(\mathbf{w}^\top \mathbf{x}) < 0 \end{cases}$$

$\hat{y} \neq y$ says $y\mathbf{w}^\top \mathbf{x} \leq 0$ (ignoring the case of 0)

The margin of a model h on an example $(\mathbf{x}, y)$ is defined as $yh(\mathbf{x})$.

- A model with a positive margin means a correct prediction
- A model with a negative margin means an incorrect prediction
- This motivates us to find a model with large margin: a large margin means the model is robust in making a correct prediction

Loss Function for Classification

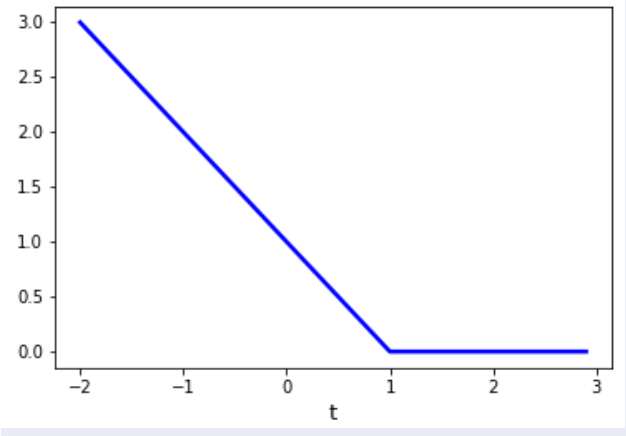
Margin-based Loss and Margin Maximization

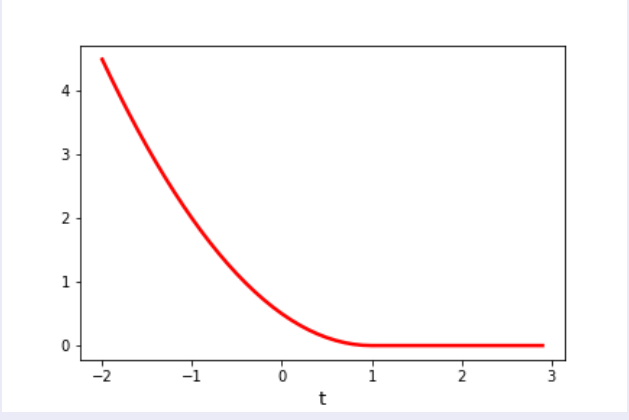
We consider the loss associated to a decreasing function $g : \mathbb{R} \mapsto \mathbb{R}$

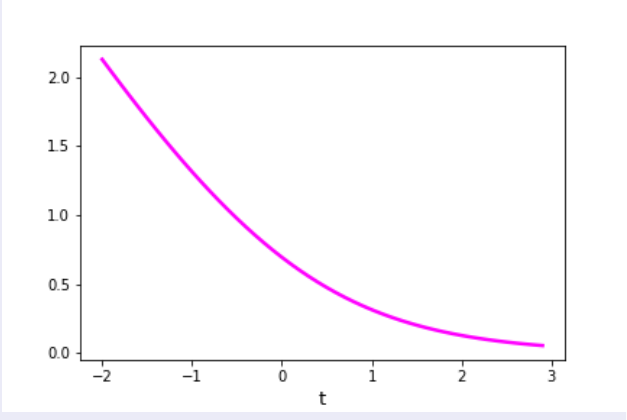
$$\ell(\hat{y}, y) = g(y\hat{y}), \quad \hat{y} = h(\mathbf{x}).$$

maximize the margin $\Longleftarrow$ minimize the margin-based loss $\ell(y, \hat{y})$

Popular Choices







$g(t) = \max\{0, 1 - t\}$ $\quad$ $g(t) = \max\{0, 1 - t\}^2/2$ $\quad$ $g(t) = \log(1 + \exp(-t))$

Empirical and Population Risk

Loss Function

We denote $f(\mathbf{w}; \mathbf{z})$ as the loss by using $h_{\mathbf{w}}$ to do prediction at $\mathbf{z}$, i.e.,

$$f(\mathbf{w}; \mathbf{z}) = \ell\big(h_{\mathbf{w}}(\mathbf{x}), y\big).$$

- e.g., for regression, $f(\mathbf{w}; \mathbf{z}) = \frac{1}{2}(h_{\mathbf{w}}(\mathbf{x}) - y)^2$
- for classification, $f(\mathbf{w}; \mathbf{z}) = \log(1 + \exp(-y h_{\mathbf{w}}(\mathbf{x})))$

Empirical and Population Risk

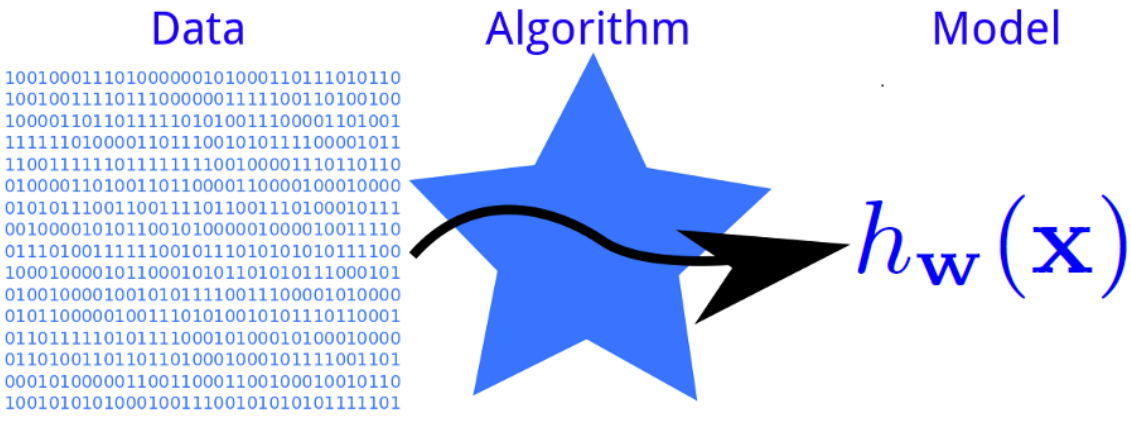
The empirical risk $F_S(\mathbf{w})$ and population risk $F(\mathbf{w})$ of a model $\mathbf{w}$ is defined by

$$F_S(\mathbf{w}) = \frac{1}{n}\sum_{i=1}^{n} f(\mathbf{w}; \mathbf{z}_i) \quad \text{and} \quad F(\mathbf{w}) = \mathbb{E}_{\mathbf{z}}[f(\mathbf{w}; \mathbf{z})]$$

- Empirical risk measures the performance on training, while population risk considers testing
- Empirical risk can be computed by the data, while population risk is in general not computable
- We often train a model based on empirical risks, while our aim is to get model with small population risks.

Algorithms

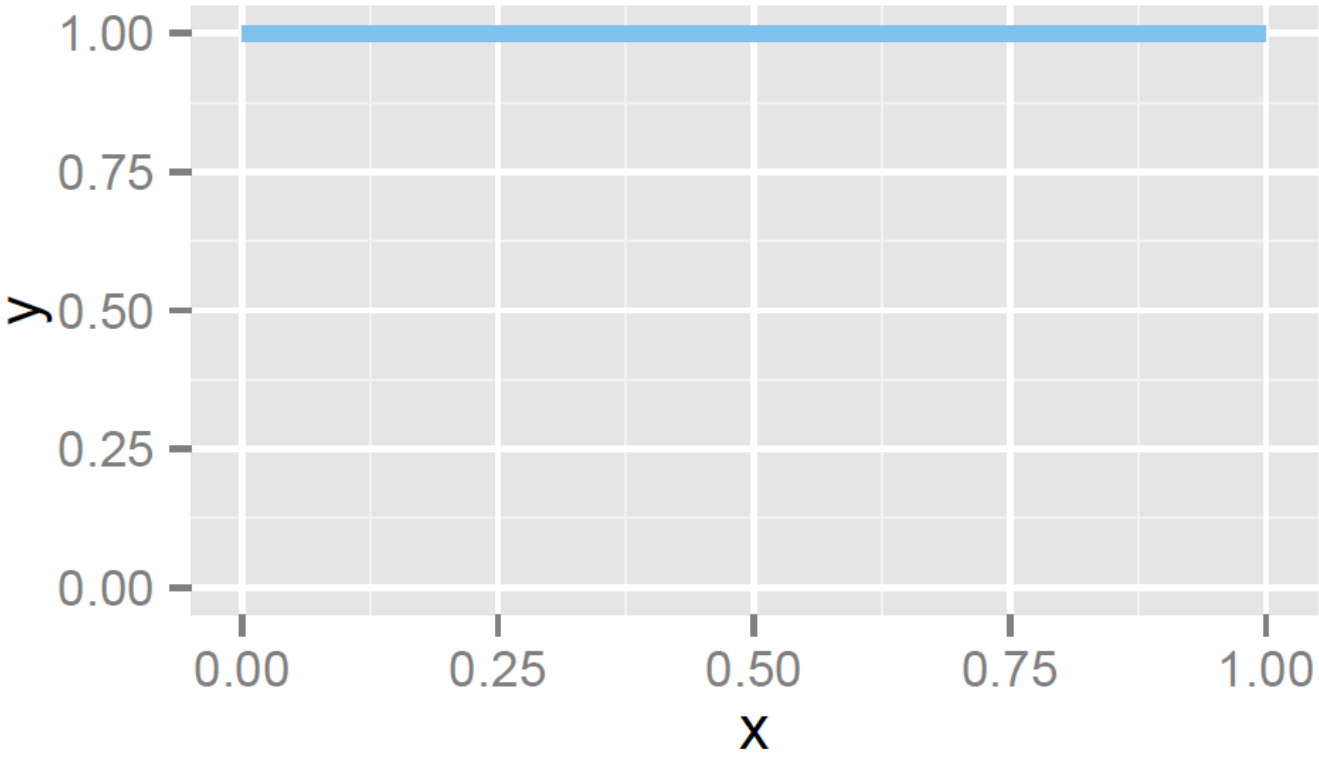
- We often apply an algorithm A to find a hypothesis $h \in \mathcal{H}$ such that h has good performance on S.
- We use $A(S)$ to denote the model derived by applying A to S



- empirical risk minimization: $A(S) = \arg\min_{\mathbf{w} \in \mathcal{W}} F_S(\mathbf{w})$
- We want risk minimizer, is empirical risk minimizer close enough? In practice, we only have a finite sample

Empirical Risk Minimization

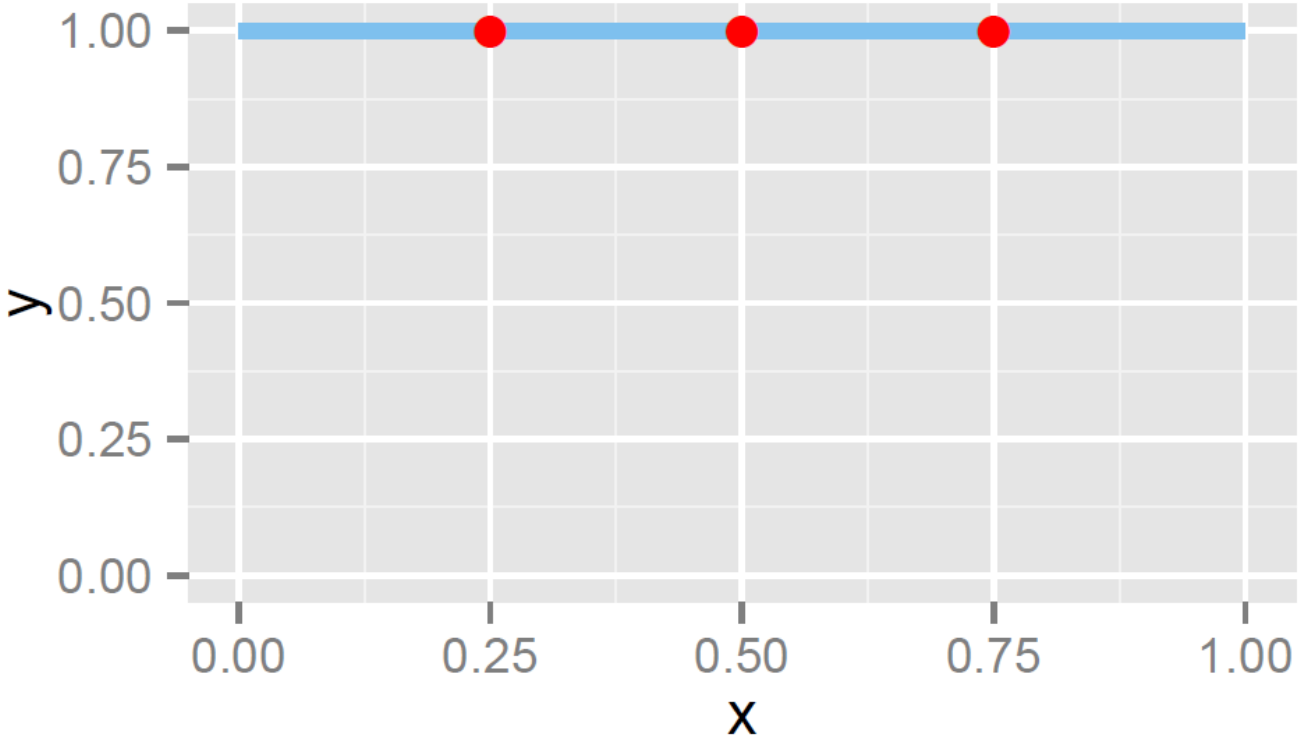
$\mathbb{P}_X = \text{Unif}[0, 1]$, $Y = 1$ (i.e., Y is always 1)



$\mathbb{P}$

Empirical Risk Minimization

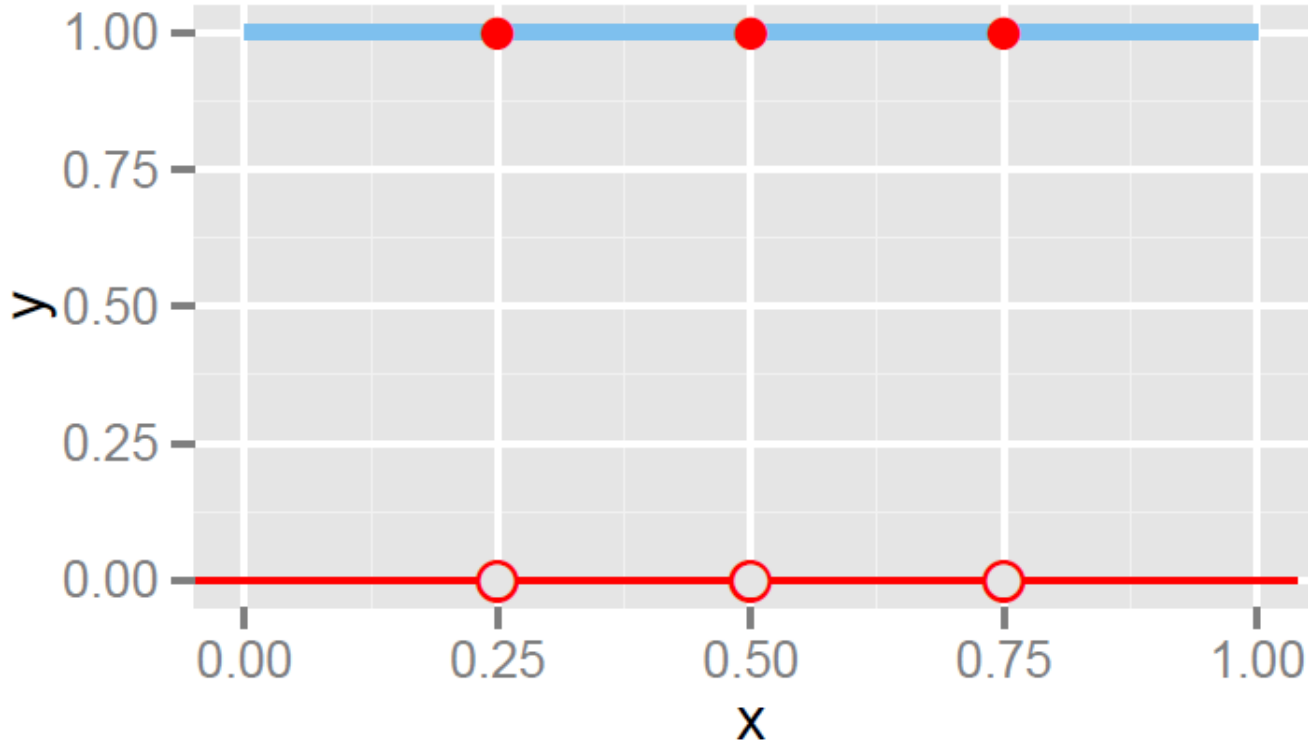
$\mathbb{P}_X = \text{Unif}[0, 1]$, $Y = 1$ (i.e., Y is always 1)



A sample of size 3 from $\mathbb{P}$

Empirical Risk Minimization

$\mathbb{P}_X = \text{Unif}[0, 1]$, $Y = 1$ (i.e., Y is always 1)



- A proposed prediction function

$$h(x) = \begin{cases} 1, & \text{if } x \in \{0.25, 0.5, 0.75\} \\ 0, & \text{otherwise.} \end{cases}$$

- Under square loss or 0/1 loss: h has empirical risk $=0$ and Risk $=1$

Other Algorithms

ERM led to a function h that just memorized the data

How to spread information or generalize from training inputs to new input?

Consider alternative algorithms

- regularized risk minimization:

$$A(S) = \arg\min_{\mathbf{w}\in\mathcal{W}} \left\{ F_S(\mathbf{w}) + \text{regularizer}(\mathbf{w}) \right\}$$

- gradient descent, stochastic gradient descent, stochastic gradient descent ascent ...

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \nabla F_S(\mathbf{w}_t),$$

  where ∇ denotes the gradient operator.

Optimization and Generalization

Excess Risk and Error Decomposition

Excess risk

The relative behavior of an output model $A(S)$ as compared to the best model $\mathbf{w}^*$ can be quantified by the excess risk $F(A(S)) - F(\mathbf{w}^*)$, where

$$\mathbf{w}^* = \arg\min_{\mathbf{w}\in\mathcal{W}} F(\mathbf{w}) \text{ is the best model.}$$

Goal: train a model as small excess risk as possible! How to estimate the excess risk?

Error decomposition

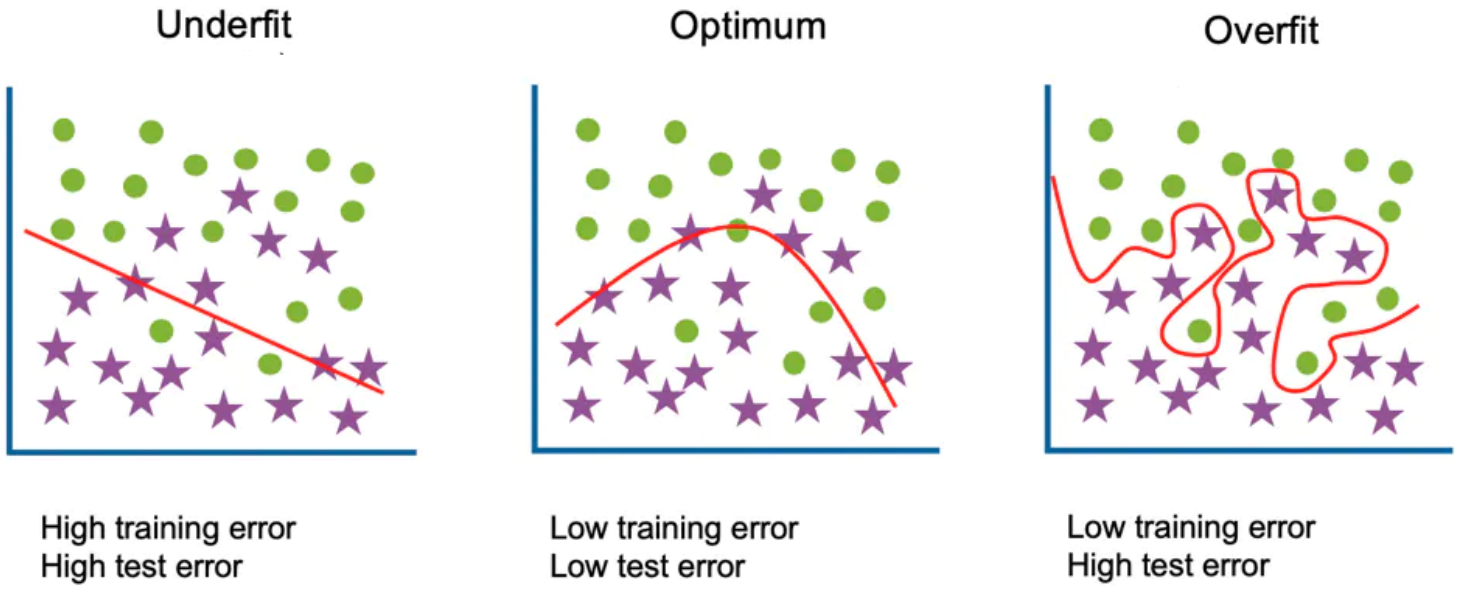
We decompose the excess risk into

$$F(A(S)) - F(\mathbf{w}^*) = \big(F(A(S)) - F_S(A(S))\big) + \big(F_S(A(S)) - F_S(\mathbf{w}^*)\big) + \big(F_S(\mathbf{w}^*) - F(\mathbf{w}^*)\big)$$

- $F(A(S)) - F_S(A(S))$: difference between training and testing at the output $A(S)$
- $F_S(A(S)) - F_S(\mathbf{w}^*)$: difference between $A(S)$ and $\mathbf{w}^*$, as measured by training error
- $F_S(\mathbf{w}^*) - F(\mathbf{w}^*)$: difference between training and testing at the best model $\mathbf{w}^*$

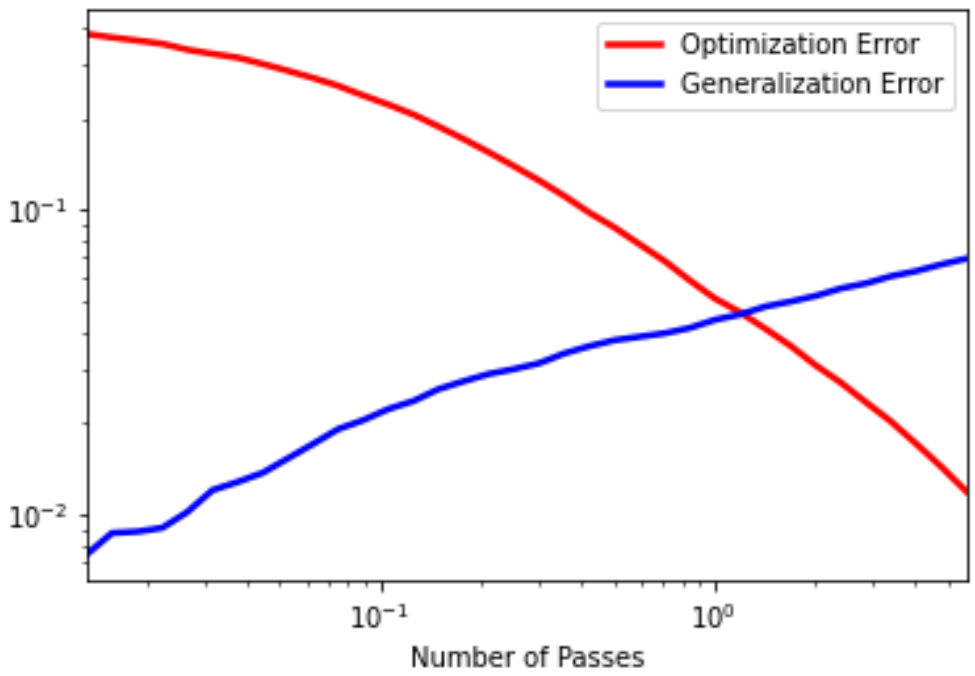
Generalization and Optimization Errors

- If the model has a large generalization gap, then the model overfits the data
- If the model has a large optimization error, then the model underfits the data



Generalization and Optimization for SGD

- Optimization errors decrease as we increase the number of iterations
- Generalization errors (gap) increase as we increase the number of iterations
- We need to balance these two errors by early-stopping



Optimization Error

We refer to $F_S(A(S)) - F_S(\mathbf{w}^*)$ as the optimization error.

- It is a topic in optimization theory
- A standard result in optimization shows: if A is the gradient descent and F_S is convex and smooth, then with appropriate step size

$$F_S(\mathbf{w}_T) - F_S(\mathbf{w}^*) \lesssim 1/(\eta T), \text{ where } \mathbf{w}_T \text{ is the } T\text{-th gradient descent iterate}$$

- Optimization error continue to decrease as we run more iterates!

Generalization Error

We refer to $F(A(S)) - F_S(A(S))$ and $F_S(\mathbf{w}^*) - F(\mathbf{w}^*)$ as the generalization error (gap).

- It is a topic in learning theory, which can be handled by tools in probability theory.
- With $\xi_i = f(\mathbf{w}^*; \mathbf{z}_i)$, then ($\mathbb{E}[\cdot]$ denotes the expectation operator)

$$F_S(\mathbf{w}^*) - F(\mathbf{w}^*) = \frac{1}{n}\sum_{i=1}^{n} f(\mathbf{w}^*; \mathbf{z}_i) - \mathbb{E}_{\mathbf{z}}[f(\mathbf{w}^*; \mathbf{z})] = \frac{1}{n}\sum_{i=1}^{n} \big[\xi_i - \mathbb{E}[\xi_i]\big].$$

  This shows that $F_S(\mathbf{w}^*) - F(\mathbf{w}^*)$ can be written as an average of independent and identically distributed (i.i.d.) random variables!

- Furthermore, we have

1.每项error不独立 2.mean不为0，即下面问题答案是：不等于

$$F(A(S)) - F_S(A(S)) = \frac{1}{n}\sum_{i=1}^{n} \Big(\mathbb{E}_{\mathbf{z}}[f(A(S); \mathbf{z})] - f(A(S); \mathbf{z}_i)\Big).$$

Is $\mathbb{E}_{\mathbf{z}}[f(A(S); \mathbf{z})]$ equal to $\mathbb{E}_{\mathbf{z}_i}[f(A(S); \mathbf{z}_i)]$?

Generalization Error

- Recall that $(\xi_i = f(\mathbf{w}^*; \mathbf{z}_i))$

$$F_S(\mathbf{w}^*) - F(\mathbf{w}^*) = \frac{1}{n} \sum_{i=1}^{n} \big[\xi_i - \mathbb{E}[\xi_i]\big].$$

  This is the difference between an empirical average and the expectation!

- Recall that

$$F(A(S)) - F_S(A(S)) = \frac{1}{n} \sum_{i=1}^{n} \Big(\mathbb{E}_{\mathbf{z}}[f(A(S); \mathbf{z})] - f(A(S); \mathbf{z}_i)\Big).$$

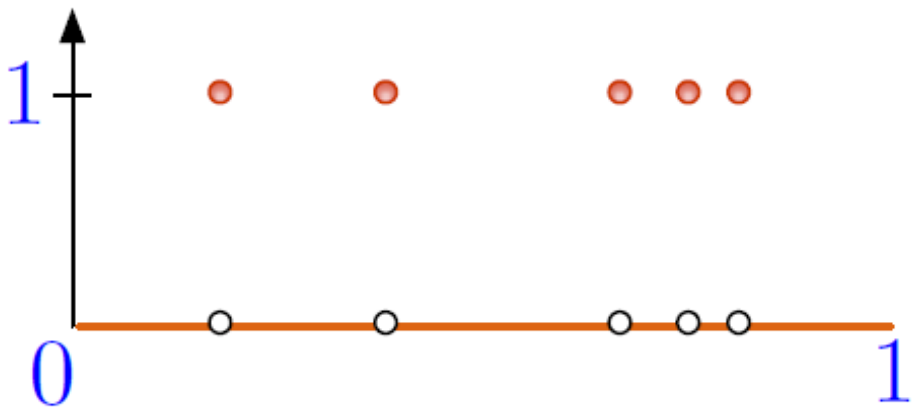
  Each summand $\mathbb{E}_{\mathbf{z}}[f(A(S); \mathbf{z})] - f(A(S); \mathbf{z}_i)$ is not mean-zero due to the bias of A!

How bad can this bias be?

Example

- $\mathcal{X} = [0,1], \mathcal{Y} = \{0,1\}$ and $\ell(h(x); y) = \mathbb{I}[h(x) \neq y]$
- $\mathbb{P}_X = \text{Unif}[0,1]$, $Y = 1$
- Function class (each h is indexed by a set $V \subseteq \mathcal{X}$, $\mathcal{H}$ is derived by traversing all V)

$$\mathcal{H} = \left\{ h_V : V \subset \mathcal{X}, h_V(x) = \mathbb{I}[x \in V] \right\}.$$



$$\hat{h}(\mathbf{x}) = \begin{cases} 1, & \text{if } \mathbf{x} \text{ is seen in } S \\ 0, & \text{otherwise.} \end{cases}$$

- ERM $\hat{h}$ memorizes (perfectly fits the data), but has no ability to generalize

$$0 = \mathbb{E}_S[\ell(\hat{h}(x_i), y_i)] \neq \mathbb{E}_{S,z}[\ell(\hat{h}(x), y)] = 1,$$

where last identity holds since $\hat{h}$ only takes value 1 for a finite number of points.

Uniform Deviation

- $A(S)$ depends on S. This dependency is addressed by taking supremum over $\mathbf{w} \in \mathcal{W}$

$$F(A(S)) - F_S(A(S)) \leq \sup_{\mathbf{w}\in\mathcal{W}} \left[\frac{1}{n}\sum_{i=1}^{n}\left(\mathbb{E}_{\mathbf{z}}[f(\mathbf{w};\mathbf{z})] - f(\mathbf{w};\mathbf{z}_i)\right)\right]$$
$$= \sup_{\mathbf{w}\in\mathcal{W}} \Big[F(\mathbf{w}) - F_S(\mathbf{w})\Big]. \tag{2}$$

- If $\mathcal{W} = \{\mathbf{w}_0\}$ consists of a single point, then clearly

$$\sup_{\mathbf{w}\in\mathcal{W}} \left[\frac{1}{n}\sum_{i=1}^{n}\left(\mathbb{E}_{\mathbf{z}}[f(\mathbf{w};\mathbf{z})] - f(\mathbf{w};\mathbf{z}_i)\right)\right] = \left[\frac{1}{n}\sum_{i=1}^{n}\left(\mathbb{E}_{\mathbf{z}}[f(\mathbf{w}_0;\mathbf{z})] - f(\mathbf{w}_0;\mathbf{z}_i)\right)\right].$$

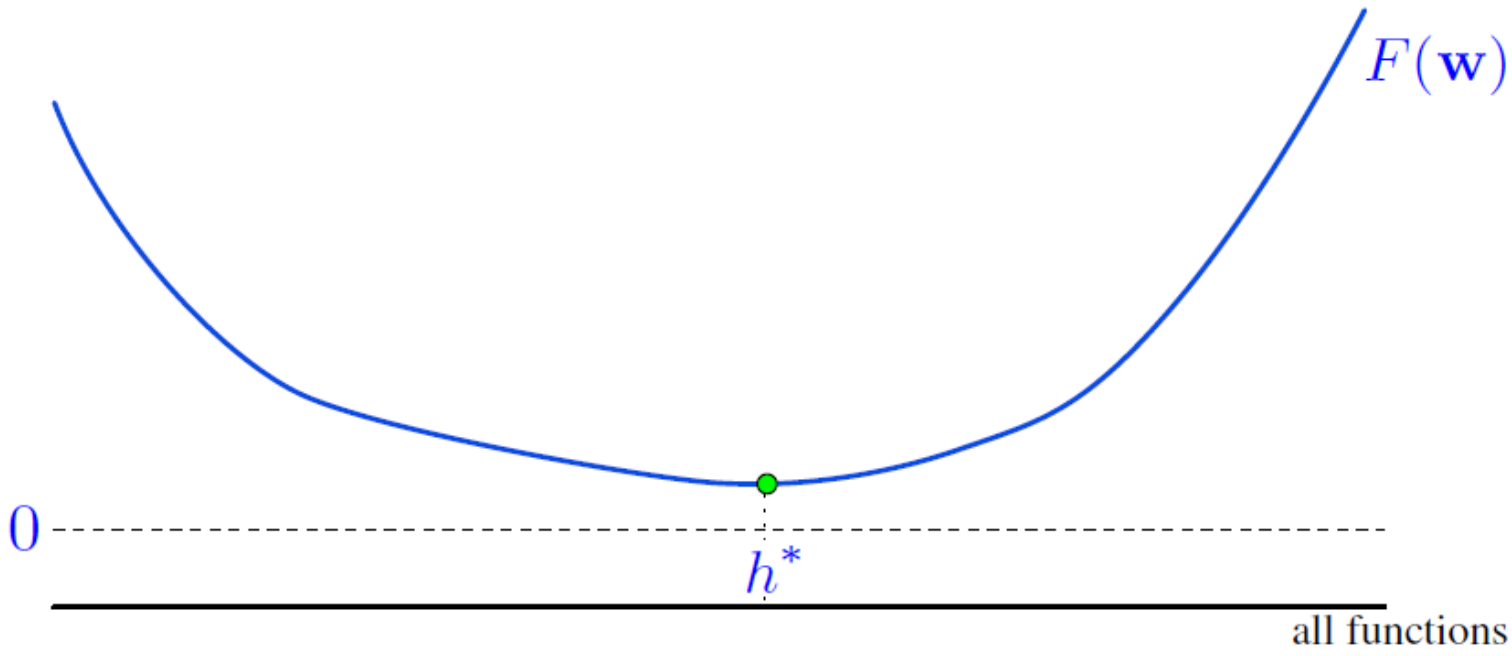
  This quantity can be easily controlled.

- The uniform deviation depends on the complexity of the hypothesis space!

Empirical Process Viewpoint

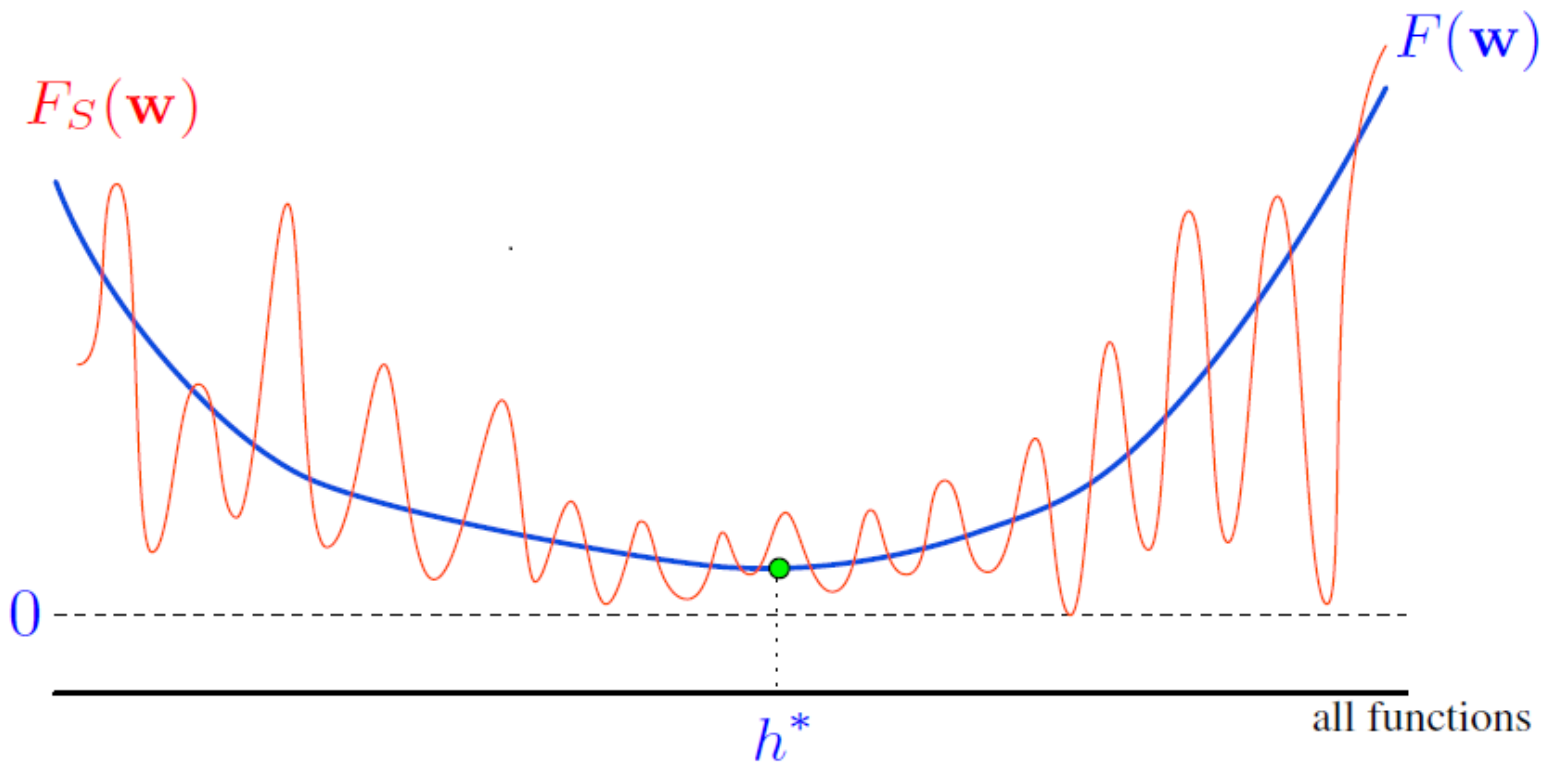
Empirical Process

For each prediction function h, the corresponding empirical risk is a random variable. If we consider all h, this builds an empirical process, i.e., a collection of random variables indexed by $\mathcal{H}$.



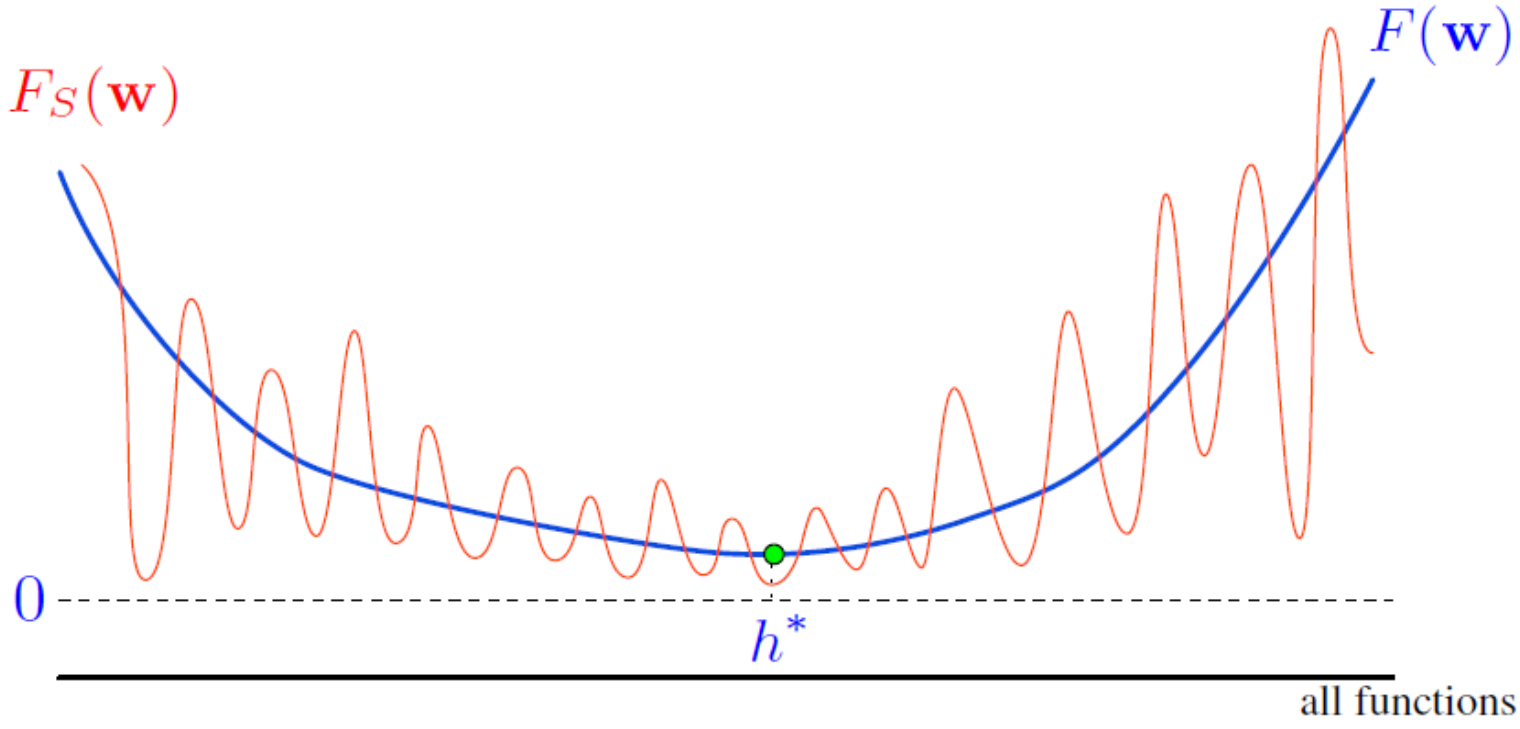
The population risk as a function of the predictors. h^* is the risk minimizer over all possible predictors, not necessarily in the hypothesis space

Empirical Process Viewpoint



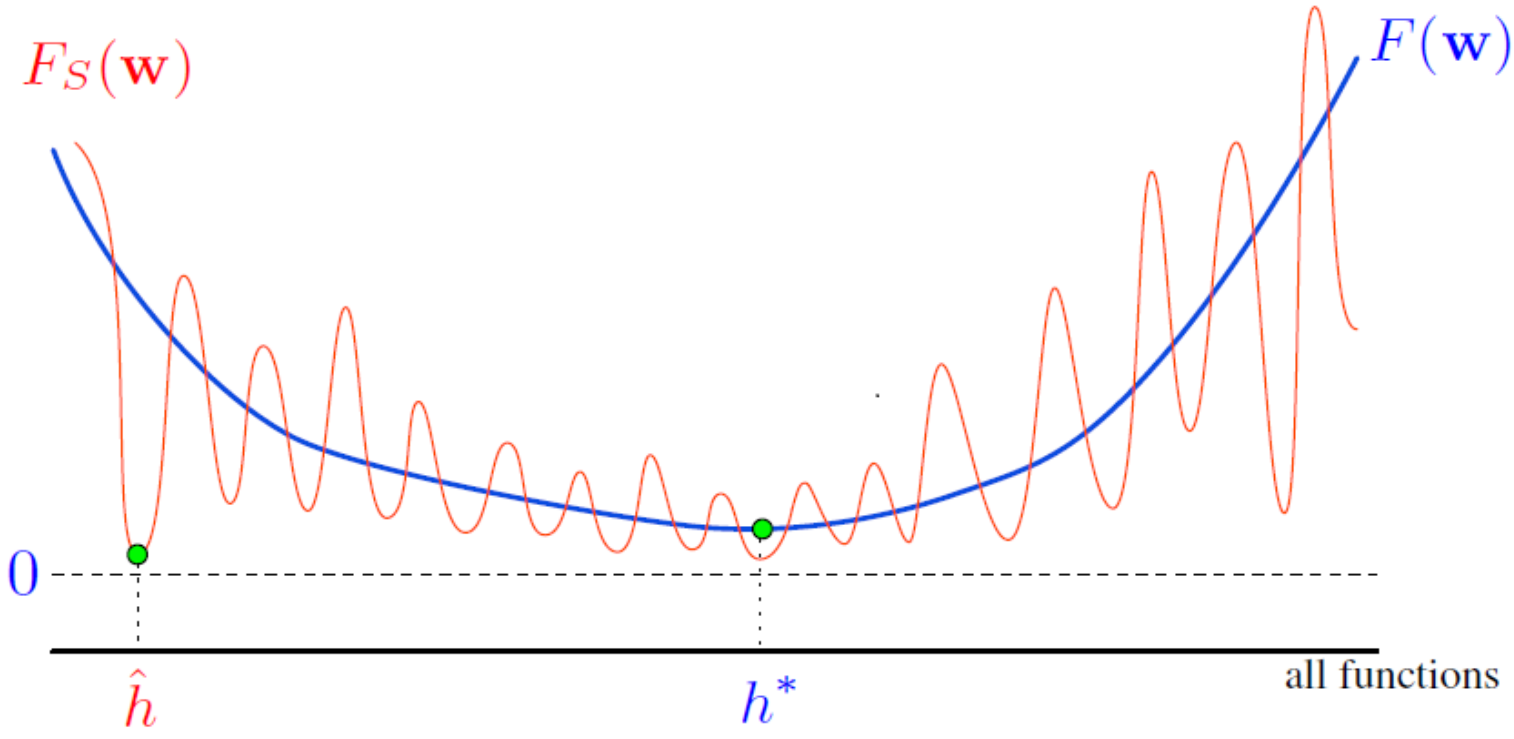
- a realization of the empirical process (i.e., a single S is drawn)

Empirical Process Viewpoint



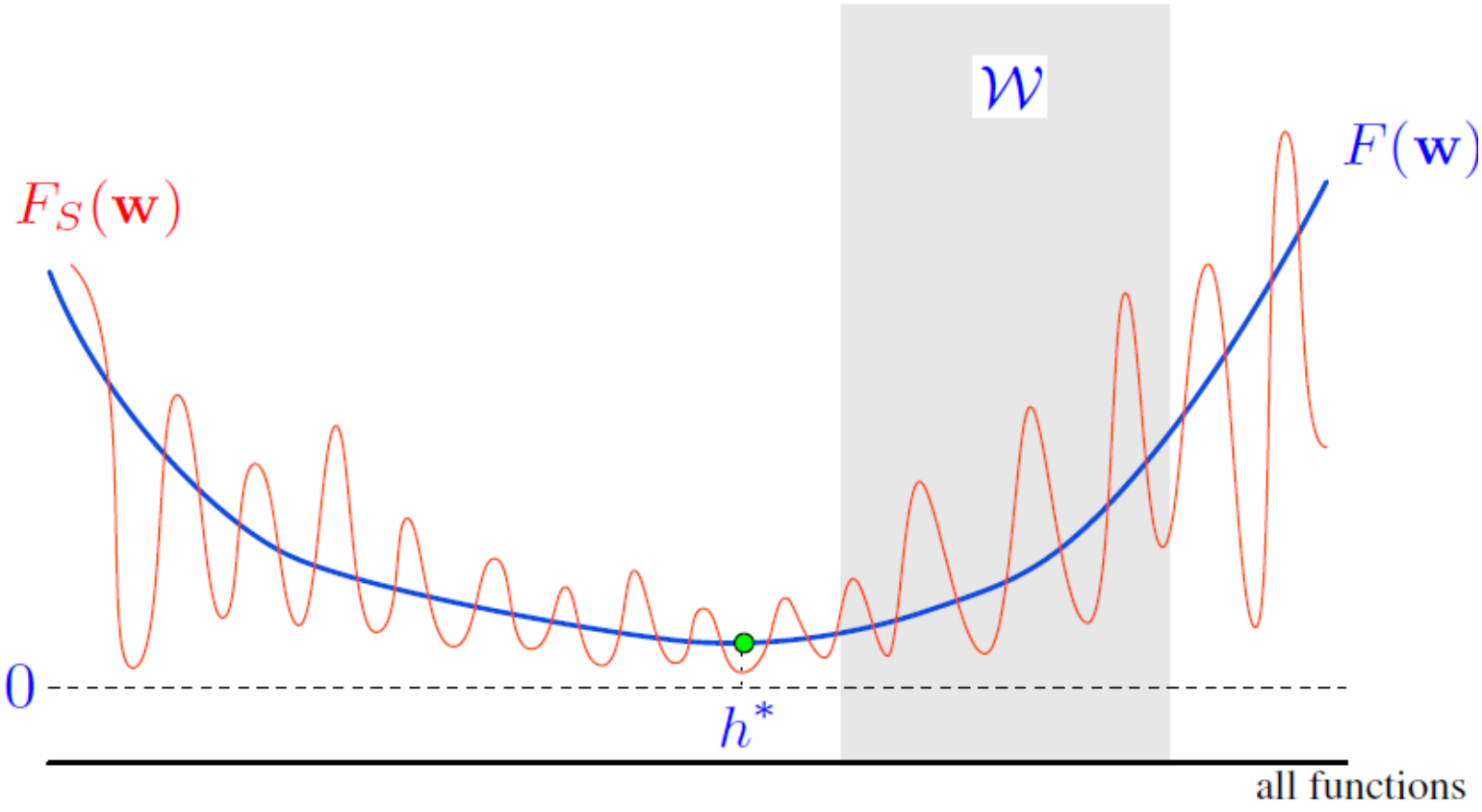
- another realization of the empirical process (i.e., another single S is drawn)

Empirical Process Viewpoint



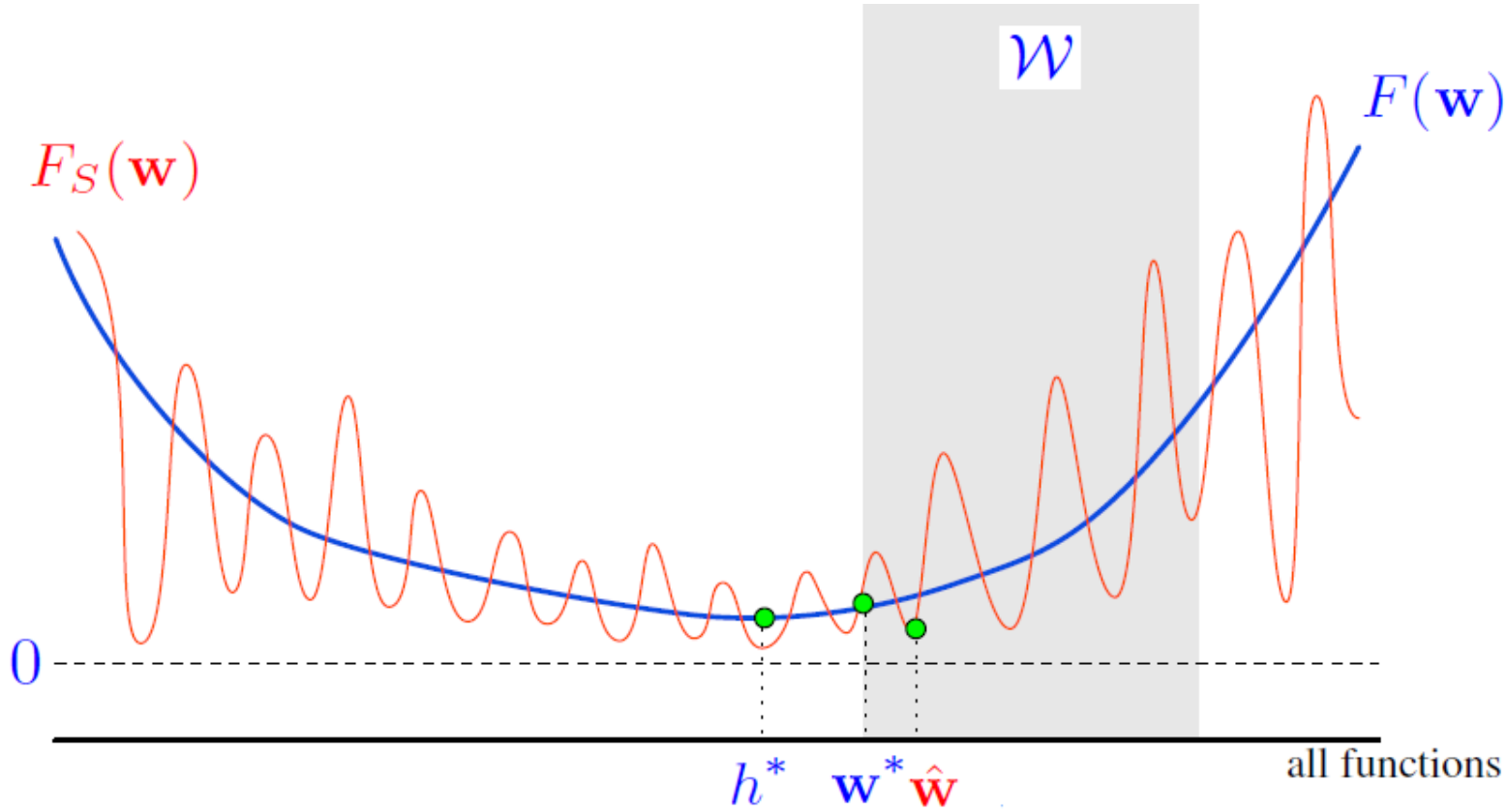
- h^* is the empirical risk minimizer over all possible predictors, not necessarily in the hypothesis space

Empirical Process Viewpoint



- Our learning is conducted in a hypothesis space, parameterized by $\mathcal{W}$

Empirical Process Viewpoint

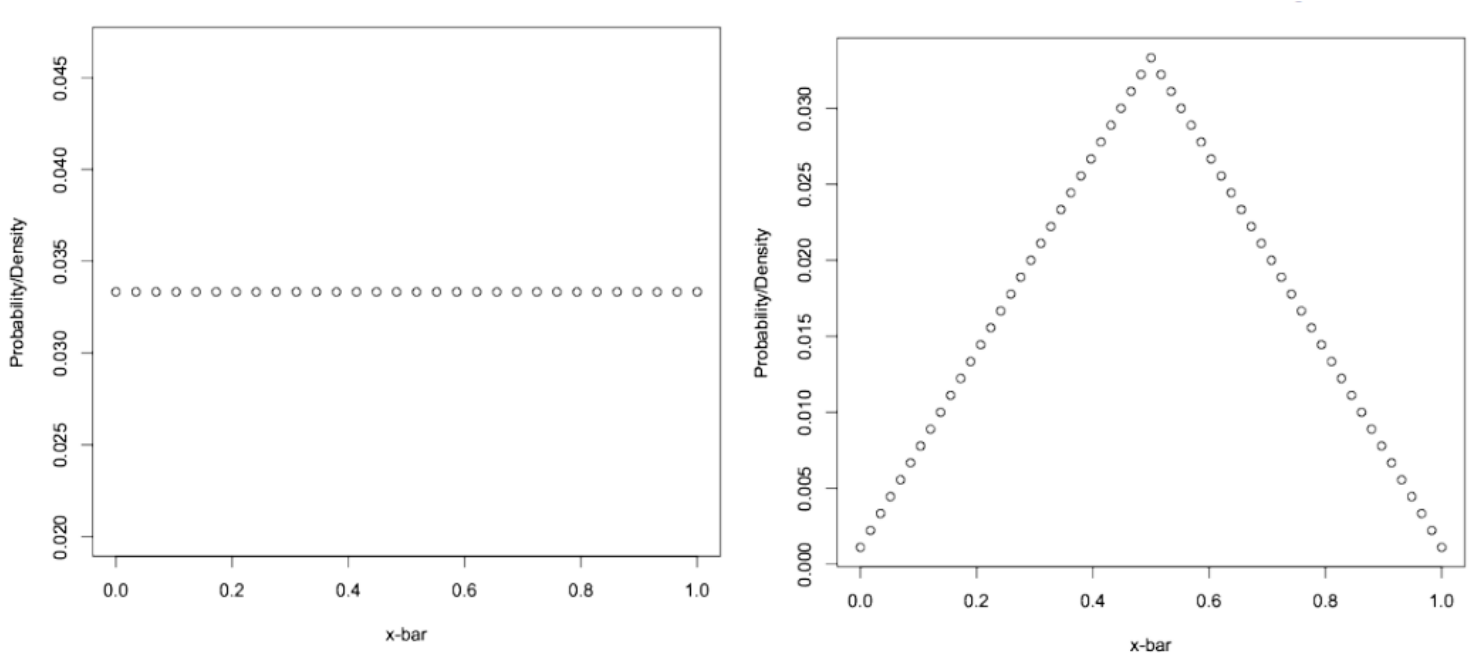


- h^* is the risk minimizer over all possible predictors, not necessarily in hypothesis space
- $\mathbf{w}^*$ is the risk minimizer over the hypothesis space ($F_S(\mathbf{w}^*)$ is an unbiased estimator of $F(\mathbf{w}^*)$)
- $\hat{\mathbf{w}}$ is the ERM over the hypothesis space ($F_S(\hat{\mathbf{w}})$ is a biased estimator of $F(\hat{\mathbf{w}})$)

Concentration Inequality

Concentration Behaviour

An intuitive example: If X_1 and X_2 are uniform in $[0, 1]$, then $\bar{X} = \frac{1}{2}(X_1 + X_2)$ is triangular



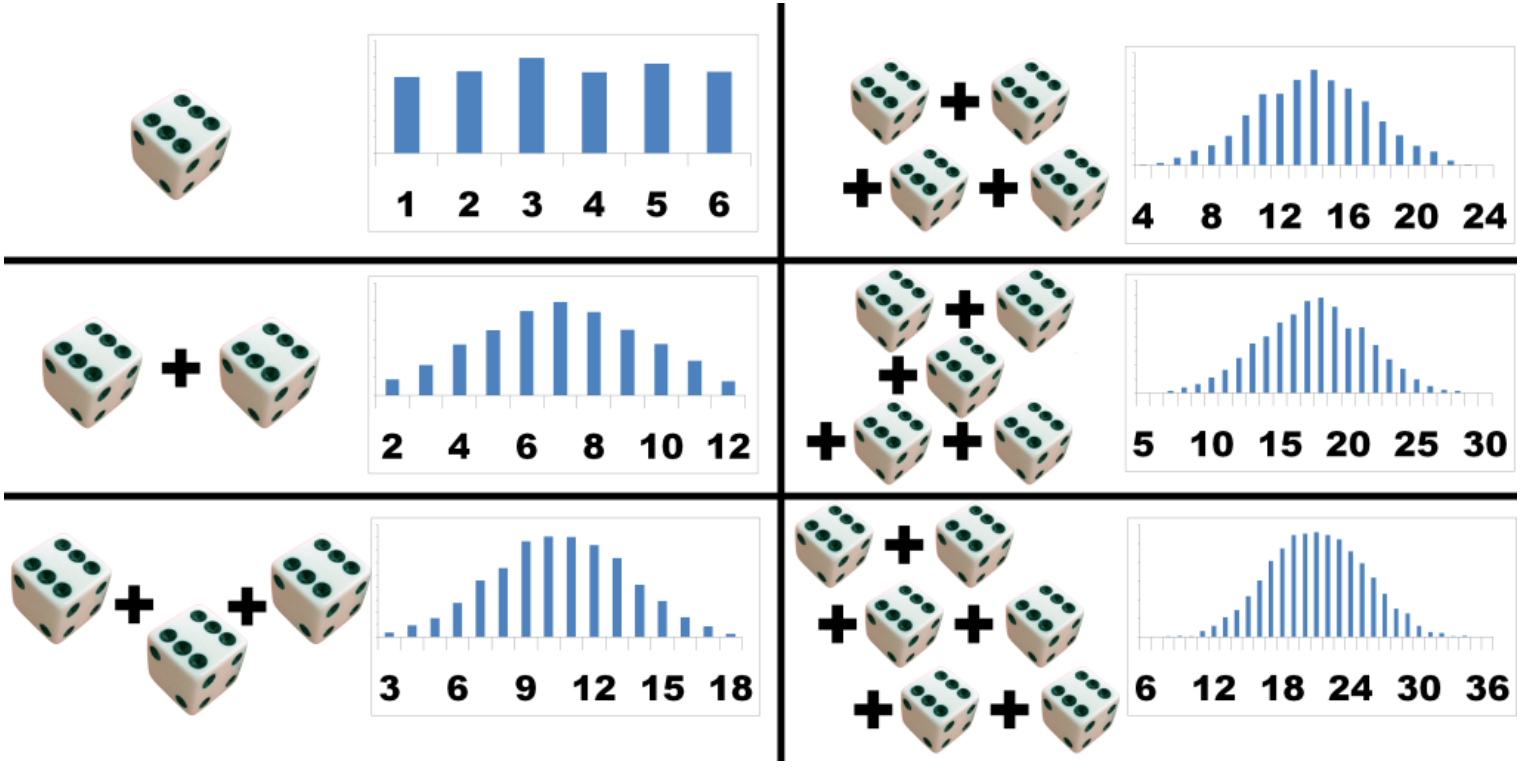
A kind of concentration property!

Central Limit Theorem

Let $X_1, \ldots, X_n$ be IID with mean μ and variance σ^2. Let $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$. Then

$$\bar{X}_n \to N(\mu, \sigma^2/n) \quad \text{as } n \to \infty,$$

where $N(\mu, \sigma^2/n)$ means the normal distribution with mean μ and variance σ^2/n.



Central Limit Theorem

Central Limit Theorem (CLT):

- CLT shows that an average of n IID random variables converges to a normal distribution
- This is asymptotical in nature: one needs n goes to infinity to get this convergence
- CLT considers a simple function: the average of random variables

Concentration inequality

- Concentration inequality also concerns a function of n random variables
- Concentration inequality is nonasymptotical: it considers a finite n
- It considers a general function g: not necessary the average
  - E.g., g is the uniform deviation of empirical process

Markov and Chebyshev Inequality

Markov's Inequality

Let X be a nonnegative random variable and $a > 0$. Then, $\mathbb{P}(X \geq a) \leq \mathbb{E}[X]/a$.

Proof. It is clear that any x can be represented by

$$x = x\mathbb{I}[x \geq a] + x\mathbb{I}[x < a].$$

Then, we know

$$\begin{aligned}\mathbb{E}[X] &= \mathbb{E}[X\mathbb{I}[x \geq a]] + \mathbb{E}[X\mathbb{I}[X < a]] \geq \mathbb{E}[X\mathbb{I}[x \geq a]] \\ &\geq \mathbb{E}[a\mathbb{I}[x \geq a]] = a\mathbb{P}(X \geq a).\end{aligned}$$

Chebyshev's Inequality

Let X be a random variable with mean μ and variance σ^2. Then, for any $a > 0$ we have

$$\mathbb{P}\big(|X - \mu| \geq a\big) \leq \sigma^2/a^2.$$

Proof. We know that

$$\mathbb{P}\big(|X - \mu| \geq a\big) = \mathbb{P}\big((X - \mu)^2 \geq a^2\big) \leq \mathbb{E}[(X - \mu)^2]/a^2 = \sigma^2/a^2.$$

Why Concentration Inequality?

Question: Toss a fair coin n times. What is the probability that we get at least $\frac{3}{4}n$ heads?



- Let $X = 1$ if we get head, and 0 otherwise
- Let S_n denote the number of heads. Then

$$\mathbb{E}[S_n] = n/2, \quad \mathrm{Var}(S_n) = n/4.$$

Why Concentration Inequality?

- By Chebyshev's inequality, we get a slow linear rate

$$\begin{aligned}\mathbb{P}\Big(S_n \geq \frac{3n}{4}\Big) &\leq \mathbb{P}\Big(\Big|S_n - \frac{n}{2}\Big| \geq \frac{n}{4}\Big) \\ &\leq \frac{\mathrm{Var}(S_n)}{(n/4)^2} = \frac{n/4}{(n/4)^2} = \frac{4}{n}.\end{aligned}$$

- One expects a faster decay by CLT. Introduce $Z_n = (S_n - n/2)/\sqrt{n/4}$.
- By CLT, Z_n converges to the standard normal distribution $N(0,1)$, and we expect

$$\mathbb{P}(S_n \geq 3n/4) = \mathbb{P}\big(Z_n \geq \sqrt{n/4}\big) \approx \mathbb{P}(g \geq \sqrt{n/4}) \leq \frac{1}{\sqrt{2\pi}} \exp(-n/8), \qquad (3)$$

where for $g \sim N(0,1)$ we know $\mathbb{P}(g \geq a) \leq \frac{1}{a}\frac{1}{\sqrt{2\pi}} \exp(-a^2/2), \quad a \geq 0.$

Why Concentration Inequality?

The analysis is based on an approximation in Eq. (3). However, this approximation admits a slow decay rate.

Berry-Esseen central limit theorem

Let X_i be i.i.d.. Then for any n and $a \in \mathbb{R}$ we have

$$\left|\mathbb{P}(Z_n \geq a) - \mathbb{P}(g \geq a)\right| \leq \rho/\sqrt{n},$$

where $\rho = \mathbb{E}[|X_1 - \mu|^3]/\sigma^3$ and $g \sim N(0, 1)$.

- We consider a more challenging problem of estimating the convergence of the distribution!
- Our problem is much simpler: the decay of Z_i.
- Concentration inequality controls this decay without resorting the convergence of distribution.

Concentration Inequality

- Intuitively, concentration inequality studies the decay of the deviation of a random variable from its expectation.
- Let $Z_1, \ldots, Z_n$ be i.i.d. random variables and $g : \mathcal{Z}^n \mapsto \mathbb{R}$. Then, we study the deviation:
$$\Delta := g(Z_1, \ldots, Z_n) - \mathbb{E}_Z[g(Z_1, \ldots, Z_n)].$$
- Let $\delta \in (0, 1)$, the desired bound on the deviation should take the following form

  with probability at least $1 - \delta$, we have $\left|g(Z_1, \ldots, Z_n) - \mathbb{E}_Z[g(Z_1, \ldots, Z_n)]\right| \leq \epsilon(\delta, n)$,

  where $\epsilon(\delta, n)$ is a function of δ and n, e.g., $\epsilon(\delta, n) = \left(\log(1/\delta)/n\right)^{\frac{1}{2}}$.
  - A simple case is $g(Z_1, \ldots, Z_n) = \frac{1}{n}\sum_{i=1}^{n} Z_i$
- There are many powerful concentration inequalities, e.g., Hoeffding inequality, Bernstein inequality, Mcdiarmid inequality...

We can use concentration inequality to control the generalization error!

Concentration Inequality for Generalization Error

Generalization error at $\mathbf{w}^*$: $F_S(\mathbf{w}^*) - F(\mathbf{w}^*) = \frac{1}{n}\sum_{i=1}^n f(\mathbf{w}^*; \mathbf{z}_i) - \mathbb{E}_{\mathbf{z}}[f(\mathbf{w}^*; \mathbf{z})]$

- If we define $g(z_1, \ldots, z_n) = \frac{1}{n}\sum_{i=1}^n f(\mathbf{w}^*; \mathbf{z}_i)$, then

$$F_S(\mathbf{w}^*) - F(\mathbf{w}^*) = g(z_1, \ldots, z_n) - \mathbb{E}_Z[g(Z_1, \ldots, Z_n)]$$

Generalization error at $A(S)$: recall that

$$F(A(S)) - F_S(A(S)) \leq \underbrace{\sup_{\mathbf{w}\in\mathcal{W}} \left[\frac{1}{n}\sum_{i=1}^n \Big(\mathbb{E}_{\mathbf{z}}[f(\mathbf{w}; \mathbf{z})] - f(\mathbf{w}; \mathbf{z}_i)\Big)\right]}_{:= g(\mathbf{z}_1, \ldots, \mathbf{z}_n)}.$$

Then, we have the following decomposition

$$F(A(S)) - F_S(A(S)) \leq \big(g(\mathbf{z}_1, \ldots, \mathbf{z}_n) - \mathbb{E}_Z[g(Z_1, \ldots, Z_n)]\big) + \mathbb{E}_Z[g(Z_1, \ldots, Z_n)].$$

- The 1st term can be handled by concentration inequality.
- The 2nd term can be handled by capacity measure of a hypothesis space.

Assumption. For simplicity, we assume $f(\mathbf{w}; \mathbf{z}) \in [0, 1]$.

Bounded difference assumption

We say a function $g : \mathcal{Z}^n \mapsto \mathbb{R}$ satisfies the bounded difference assumption if there exist $c_1, \ldots, c_n > 0$ such that

$$\sup_{\mathbf{z}_1, \ldots, \mathbf{z}_n, \mathbf{z}_i'} \left| g(\mathbf{z}_1, \ldots, \mathbf{z}_n) - g(\mathbf{z}_1, \ldots, \mathbf{z}_{i-1}, \mathbf{z}_i', \mathbf{z}_{i+1}, \ldots, \mathbf{z}_n) \right| \leq c_i, \quad \forall i \in [n].$$

McDiarmid's Inequality

Let $Z_1, \ldots, Z_n$ be independent random variables. If $g : \mathcal{Z}^n \mapsto \mathbb{R}$ satisfies the bounded difference assumption, then with probability at least $1 - \delta$ we have

$$g(Z_1, \ldots, Z_n) \leq \mathbb{E}_Z[g(Z_1, \ldots, Z_n)] + \left(\frac{\log(1/\delta)}{2} \sum_{i=1}^{n} c_i^2 \right)^{\frac{1}{2}}.$$

Similarly, the following inequality also holds with probability at least $1 - \delta$

$$\mathbb{E}_Z[g(Z_1, \ldots, Z_n)] \leq g(Z_1, \ldots, Z_n) + \left(\frac{\log(1/\delta)}{2} \sum_{i=1}^{n} c_i^2 \right)^{\frac{1}{2}}.$$

If a change of any single argument leads to a small change c_i, then the random variable $g(Z_1, \ldots, Z_n)$ concentrates around its expectation!

Proof of McDiarmid's Inequality [Optional]

Markov's Inequality

For a non-negative random variable, for any $t > 0$ we have

$$\Pr(X \geq t) \leq \mathbb{E}[X]/t$$

Hoeffding's Lemma

Let X be a mean-zero random variable with $a \leq X \leq b$. Then for $t > 0$

$$\mathbb{E}[\exp(tX)] \leq \exp\left(\frac{t^2(b-a)^2}{8}\right).$$

Proof of McDiarmid's Inequality [Optional]

- Let $\mathbf{Z}_1^i$ be the sequence of random variables $Z_1, \ldots, Z_i$.
- Denote $X_i = \mathbb{E}[g(\mathbf{Z})|\mathbf{Z}_1^i]$, where $\mathbf{Z} = (Z_1, \ldots, Z_n)$.
- Observe that $X_0 = \mathbb{E}[g(\mathbf{Z})]$ and $X_n = g(\mathbf{Z})$. Then

$$g(\mathbf{Z}) - \mathbb{E}[g(\mathbf{Z})] = X_n - X_0 = \sum_{k=0}^{n-1} (X_{k+1} - X_k). \tag{4}$$

- Consider the random variable $X_i - X_{i-1}|\mathbf{Z}_1^{i-1}$

Observation 1: $\mathbb{E}[X_i - X_{i-1}|\mathbf{Z}_1^{i-1}] = 0$.
Observation 2: conditioned on $\mathbf{Z}_1^{i-1}$, we have $X_i - X_{i-1} \leq c_i$. Hoeffding's Lemma shows

$$\mathbb{E}\left[\exp(t(X_i - X_{i-1}))|\mathbf{Z}_1^{i-1}\right] \leq \exp(t^2 c_i^2/8).$$

Proof of McDiarmid's Inequality [Optional]

$$\Pr\big(g - \mathbb{E}[g] \geq \epsilon\big) = \Pr\Big(\exp(t(g - \mathbb{E}[g])) \geq \exp(t\epsilon)\Big)$$

$$\leq \exp(-t\epsilon)\mathbb{E}\big[\exp(t(g - \mathbb{E}[g]))\big] \qquad \text{Markov's inequality}$$

$$= \exp(-t\epsilon)\mathbb{E}\Big[\exp\Big(t\sum_{i=1}^{n}(X_i - X_{i-1})\Big)\Big] \qquad \text{Eq. (4)}$$

$$= \exp(-t\epsilon)\mathbb{E}\Big[\mathbb{E}\Big[\exp\Big(t\sum_{i=1}^{n}(X_i - X_{i-1})\Big)|\mathbf{Z}_1^{n-1}\Big]\Big] \qquad \text{iterative expectation}$$

$$= \exp(-t\epsilon)\mathbb{E}\Big[\exp\Big(t\sum_{i=1}^{n}(X_i - X_{i-1})\Big)\mathbb{E}\Big[\exp\Big(t(X_n - X_{n-1})\Big)|\mathbf{Z}_1^{n-1}\Big]\Big]$$

$$\leq \exp(-t\epsilon)\exp\Big(\frac{t^2c_n^2}{8}\Big)\mathbb{E}\Big[\exp\Big(t\sum_{i=1}^{n-1}(X_i - X_{i-1})\Big)\Big]$$

Just showed $\mathbb{E}\Big[\exp\Big(t\sum_{i=1}^{n}(X_i - X_{i-1})\Big)\Big] \leq \exp\Big(\frac{t^2c_n^2}{8}\Big)\mathbb{E}\Big[\exp\Big(t\sum_{i=1}^{n-1}(X_i - X_{i-1})\Big)\Big],$

We continue this way and get (we also get $\mathbb{E}\big[t(g - \mathbb{E}[g])\big] \leq \exp\Big(\sum_{i=1}^{n}\frac{t^2c_i^2}{8}\Big)$)

$$\Pr\big(g - \mathbb{E}[g]\big) \leq \exp\Big(-t\epsilon + \frac{t^2}{8}\sum_{i}^{n}c_i^2\Big).$$

Proof of McDiarmid's Inequality [Optional]

We just derived

$$\Pr\big(g - \mathbb{E}[g] \geq \epsilon\big) \leq \exp\Big(- t\epsilon + \frac{t^2}{8} \sum_{i=1}^{n} c_i^2 \Big).$$

- Choose t that minimizes $-t\epsilon + \frac{t^2}{8} \sum_{i=1}^{n} c_i^2$
- This leads to $t = \frac{4\epsilon}{\sum_{i=1}^{n} c_i^2}$ and

$$-t\epsilon + \frac{t^2}{8} \sum_{i=1}^{n} c_i^2 = \frac{-2\epsilon^2}{\sum_{i=1}^{n} c_i^2}$$

This gives

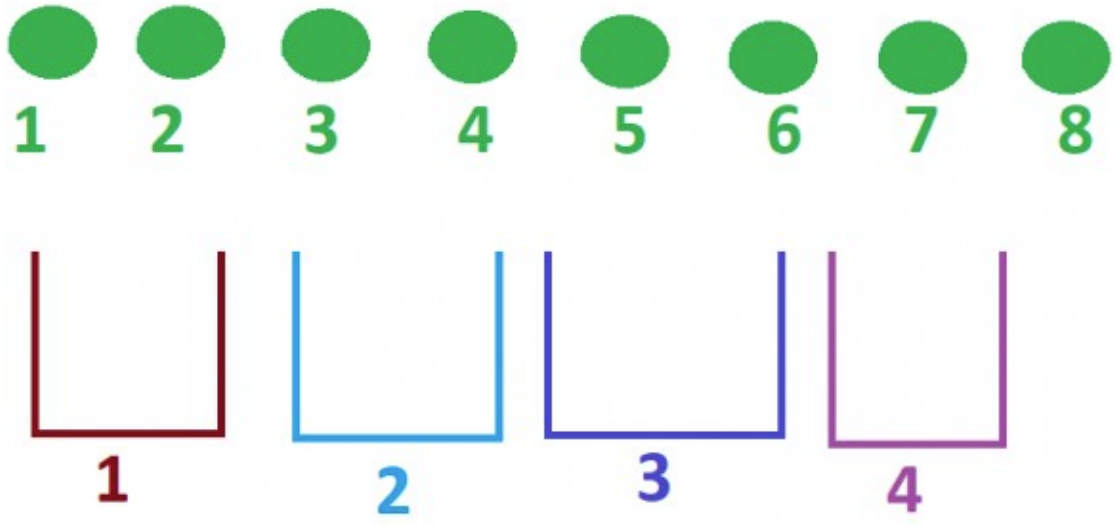
$$\Pr\big(g - \mathbb{E}[g] \geq \epsilon\big) \leq \exp\Big(- \frac{2\epsilon^2}{\sum_{i=1}^{n} c_i^2} \Big).$$

Putting $\delta = \exp\Big(- \frac{2\epsilon^2}{\sum_{i=1}^{n} c_i^2} \Big)$, we get

$$\log(1/\delta) = \frac{2\epsilon^2}{\sum_{i=1}^{n} c_i^2} \iff \epsilon = \Big(\frac{\log(1/\delta)}{2} \sum_{i=1}^{n} c_i^2 \Big)^{\frac{1}{2}}.$$

Application of McDiarmid's Inequality (Balls into Bins)

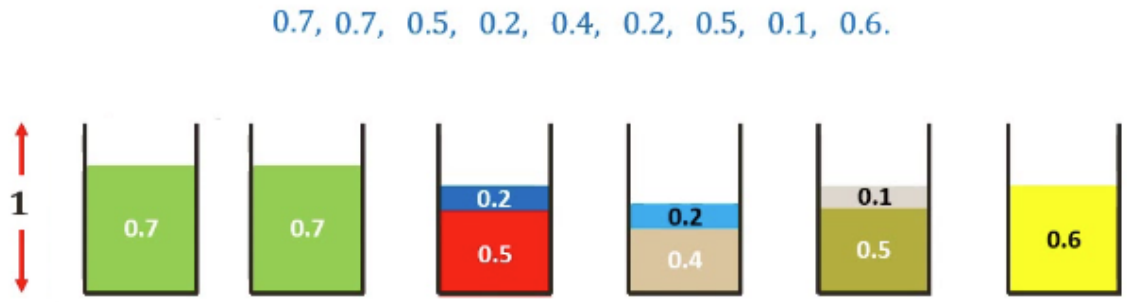
1. Suppose we have n balls assigned uniformly at random into m bins.
2. Let X_i be the bin assigned to i-th ball. Let Z be the number of empty bins.



3. Z is a function of $X_1, \ldots, X_n$, i.e.g, we can write $Z = g(X_1, \ldots, X_n)$.
4. We can show that g satisfies the bounded difference inequality with $c_i = 1$
   - Indeed, if we move one ball to another bin, the number of empty bins changes at most by 1.
   - With probability at least $1 - \delta$, $|\mathbb{E}[Z] - Z| \leq \left(2^{-1} n \log(1/\delta)\right)^{\frac{1}{2}}$!
5. The probability that the j-th bin is empty is $\left(1 - 1/m\right)^n$.

Application of McDiarmid's Inequality (Bin Packing)

1. Suppose we have n items and let X_i be the size of the i-th item.
2. We want to pack these items into the fewest number of unit-capacity bins as possible.



3. Assume X_i are independent random variables taking values uniformly from $[0, 1]$
4. Let $Z = g(X_1, \ldots, X_n)$ be the minimal number of bins that suffices to pack these items.
5. We can show that g satisfies the bounded difference inequality with $c_i = 1$
   - Indeed, if we change the size of any i-th item, the minimal number bins changes at most by 1.
   - With probability at least $1 - \delta$, $|\mathbb{E}[Z] - Z| \leq \left(2^{-1} n \log(1/\delta)\right)^{\frac{1}{2}}$!

Application of McDiarmid's Inequality (Generalization)

Hoeffding's Inequality. Let $Z_1, \ldots, Z_n$ be independent random variables with $Z_i \in [b_1, b_2]$. Then, with probability at least $1 - \delta$ we have

$$\frac{1}{n}\sum_{i=1}^{n}\left(Z_i - \mathbb{E}[Z_i]\right) \leq \frac{(b_2 - b_1)\log^{\frac{1}{2}}(1/\delta)}{\sqrt{2n}}.$$

- Define $g(Z_1, \ldots, Z_n) = \frac{1}{n}\sum_{i=1}^{n} Z_i$
- Then we have the bounded difference assumption

$$g(\mathbf{z}_1, \ldots, \mathbf{z}_n) - g(\mathbf{z}_1, \ldots, \mathbf{z}_{i-1}, \mathbf{z}_i', \mathbf{z}_{i+1}, \ldots, \mathbf{z}_n) = \frac{1}{n}\sum_{j=1}^{n}\mathbf{z}_j - \frac{1}{n}\sum_{j:j\neq i}\mathbf{z}_j - \frac{1}{n}\mathbf{z}_i'$$

$$= \frac{1}{n}(\mathbf{z}_i - \mathbf{z}_i') \leq \frac{(b_2 - b_1)}{n} := c_i.$$

- By Mcdiarmid's inequality, with probability at least $1 - \delta$

$$\frac{1}{n}\sum_{i=1}^{n}\left(Z_i - \mathbb{E}[Z_i]\right) \leq \frac{b_2 - b_1}{n}\left(\frac{n\log(1/\delta)}{2}\right)^{\frac{1}{2}} = \frac{(b_2 - b_1)\log^{\frac{1}{2}}(1/\delta)}{\sqrt{2n}}.$$

Bound on $F_S(\mathbf{w}^*) - F(\mathbf{w}^*)$: with probability at least $1 - \delta/2$, we have

$$F_S(\mathbf{w}^*) - F(\mathbf{w}^*) \leq (2n)^{-\frac{1}{2}}\log^{\frac{1}{2}}(2/\delta). \tag{5}$$

Application of McDiarmid's Inequality

We now consider $g(\mathbf{z}_1, \ldots, \mathbf{z}_n) - \mathbb{E}_Z[g(Z_1, \ldots, Z_n)]$, where

$$g(\mathbf{z}_1, \ldots, \mathbf{z}_n) = \sup_{\mathbf{w} \in \mathcal{W}} \left[\frac{1}{n} \sum_{i=1}^{n} \Big(\mathbb{E}_{\mathbf{z}}[f(\mathbf{w}; \mathbf{z})] - f(\mathbf{w}; \mathbf{z}_i) \Big) \right].$$

Lemma. For any $h_1, h_2 \in \mathcal{W}$, we have

$$\Big| \sup_{\mathbf{w} \in \mathcal{W}} h_1(\mathbf{w}) - \sup_{\mathbf{w} \in \mathcal{W}} h_2(\mathbf{w}) \Big| \leq \sup_{\mathbf{w} \in \mathcal{W}} \big| h_1(\mathbf{w}) - h_2(\mathbf{w}) \big|. \tag{6}$$

We now show g satisfies the bounded difference assumption by Eq. (6)

$$ng(\mathbf{z}_1, \ldots, \mathbf{z}_n) - ng(\mathbf{z}_1, \ldots, \mathbf{z}_{i-1}, \mathbf{z}_i', \mathbf{z}_{i+1}, \ldots, \mathbf{z}_n)$$

$$= \sup_{\mathbf{w} \in \mathcal{W}} \left[\sum_{j \in [n]} \Big(\mathbb{E}_{\mathbf{z}}[f(\mathbf{w}; \mathbf{z})] - f(\mathbf{w}; \mathbf{z}_j) \Big) \right] - \sup_{\mathbf{w} \in \mathcal{W}} \left[\sum_{j \in [n]: j \neq i} \Big(\mathbb{E}_{\mathbf{z}}[f(\mathbf{w}; \mathbf{z})] - f(\mathbf{w}; \mathbf{z}_j) \Big) + \big(\mathbb{E}_{\mathbf{z}}[f(\mathbf{w}; \mathbf{z})] -$$

$$\leq \sup_{\mathbf{w} \in \mathcal{W}} \left[\sum_{j \in [n]: j \neq i} \Big(\mathbb{E}_{\mathbf{z}}[f(\mathbf{w}; \mathbf{z})] - f(\mathbf{w}; \mathbf{z}_j) - \mathbb{E}_{\mathbf{z}}[f(\mathbf{w}; \mathbf{z})] + f(\mathbf{w}; \mathbf{z}_j) \Big) + \right.$$

$$\left. \big(\mathbb{E}_{\mathbf{z}}[f(\mathbf{w}; \mathbf{z})] - f(\mathbf{w}; \mathbf{z}_i) - \mathbb{E}_{\mathbf{z}}[f(\mathbf{w}; \mathbf{z})] + f(\mathbf{w}; \mathbf{z}_i') \big) \right] = \sup_{\mathbf{w} \in \mathcal{W}} \big[f(\mathbf{w}; \mathbf{z}_i') - f(\mathbf{w}; \mathbf{z}_i) \big] \leq 1.$$

Bound on $F(A(S)) - F_S(A(S))$: with probability at least $1 - \delta/2$, we have

$$F(A(S)) - F_S(A(S)) \leq \mathbb{E}\Big[\sup_{\mathbf{w} \in \mathcal{W}} \frac{1}{n} \sum_{i=1}^{n} \Big(\mathbb{E}_{\mathbf{z}}[f(\mathbf{w}; \mathbf{z})] - f(\mathbf{w}; \mathbf{z}_i) \Big) \Big] + \frac{1}{\sqrt{2n}} \log^{\frac{1}{2}}(2/\delta).$$

Empirical Risk Minimization

- Let us consider the empirical risk minimization algorithm

$$A(S) = \arg\min_{\mathbf{w}\in\mathcal{W}} F_S(\mathbf{w}).$$

- In this case, we have $F_S(A(S)) - F_S(\mathbf{w}^*) \leq 0$ and there is no need to consider optimization error, i.e.,

$$F(A(S)) - F(\mathbf{w}^*) \leq \big(F(A(S)) - F_S(A(S))\big) + \big(F_S(\mathbf{w}^*) - F(\mathbf{w}^*)\big)$$

- We showed with probability at least $1 - \delta/2$, $F_S(\mathbf{w}^*) - F(\mathbf{w}^*) \leq (2n)^{-\frac{1}{2}} \log^{\frac{1}{2}}(2/\delta)$
- We just showed with probability at least $1 - \delta/2$,

$$F(A(S)) - F_S(A(S)) \leq \mathbb{E}\Big[\sup_{\mathbf{w}\in\mathcal{W}} \frac{1}{n}\sum_{i=1}^{n}\Big(\mathbb{E}_{\mathbf{z}}[f(\mathbf{w};\mathbf{z})] - f(\mathbf{w};\mathbf{z}_i)\Big)\Big] + \frac{1}{\sqrt{2n}}\log^{\frac{1}{2}}(2/\delta).$$

Putting together, we get with probability at least $1 - \delta$

$$F(A(S)) - F(\mathbf{w}^*) \leq \sqrt{2}n^{-\frac{1}{2}}\log^{\frac{1}{2}}(2/\delta) + \mathbb{E}\Big[\sup_{\mathbf{w}\in\mathcal{W}} \frac{1}{n}\sum_{i=1}^{n}\Big(\mathbb{E}_{\mathbf{z}}[f(\mathbf{w};\mathbf{z})] - f(\mathbf{w};\mathbf{z}_i)\Big)\Big]. \quad (7)$$

How to control the expectation term?

Complexity Measure

Rademacher Complexity

Rademacher Complexity

- We want to control $\mathbb{E}\Big[\sup_{\mathbf{w}\in\mathcal{W}} \frac{1}{n}\sum_{i=1}^{n}\Big(\mathbb{E}_{\mathbf{z}}[f(\mathbf{w};\mathbf{z})] - f(\mathbf{w};\mathbf{z}_i)\Big)\Big]$
- Symmetry arguments: Let $S' = \{\mathbf{z}'_1, \ldots, \mathbf{z}'_n\}$ be a new i.i.d. sample. Then

$$\mathbb{E}\Big[\sup_{\mathbf{w}\in\mathcal{W}} \frac{1}{n}\sum_{i=1}^{n}\Big(\mathbb{E}_{\mathbf{z}}[f(\mathbf{w};\mathbf{z})] - f(\mathbf{w};\mathbf{z}_i)\Big)\Big] = \mathbb{E}\Big[\sup_{\mathbf{w}\in\mathcal{W}} \frac{1}{n}\sum_{i=1}^{n}\Big(\mathbb{E}_{\mathbf{z}'_i}[f(\mathbf{w};\mathbf{z}'_i)] - f(\mathbf{w};\mathbf{z}_i)\Big)\Big]$$

$$\leq \mathbb{E}_{S,S'}\Big[\sup_{\mathbf{w}\in\mathcal{W}} \frac{1}{n}\sum_{i=1}^{n}\Big(f(\mathbf{w};\mathbf{z}'_i) - f(\mathbf{w};\mathbf{z}_i)\Big)\Big]$$

where we use the fact that $\sup_{\mathbf{w}} \mathbb{E}_{S'}[g(\mathbf{w};S')] \leq \mathbb{E}_{S'}[\sup_{\mathbf{w}} g(\mathbf{w};S')]$.

- Due to the symmetry between $\mathbf{z}_i$ and $\mathbf{z}'_i$, $f(\mathbf{w};\mathbf{z}'_i) - f(\mathbf{w};\mathbf{z}_i)$ has the same distribution as $\epsilon_i\big(f(\mathbf{w};\mathbf{z}'_i) - f(\mathbf{w};\mathbf{z}_i)\big)$, where $\Pr(\epsilon_i = 1) = \Pr(\epsilon_i = -1) = 1/2$.

$$\Longrightarrow \mathbb{E}\Big[\sup_{\mathbf{w}\in\mathcal{W}} \frac{1}{n}\sum_{i=1}^{n}\Big(\mathbb{E}_{\mathbf{z}}[f(\mathbf{w};\mathbf{z})] - f(\mathbf{w};\mathbf{z}_i)\Big)\Big] \leq \mathbb{E}_{S,S',\epsilon}\Big[\sup_{\mathbf{w}\in\mathcal{W}} \frac{1}{n}\sum_{i=1}^{n}\epsilon_i\Big(f(\mathbf{w};\mathbf{z}'_i) - f(\mathbf{w};\mathbf{z}_i)$$

$$\leq \mathbb{E}_{S,S',\epsilon}\Big[\sup_{\mathbf{w}\in\mathcal{W}} \frac{1}{n}\sum_{i=1}^{n}\epsilon_i f(\mathbf{w};\mathbf{z}'_i)\Big] + \mathbb{E}_{S,S',\epsilon}\Big[\sup_{\mathbf{w}\in\mathcal{W}} \frac{1}{n}\sum_{i=1}^{n} -\epsilon_i f(\mathbf{w};\mathbf{z}_i)\Big]$$

$$= \frac{2}{n}\mathbb{E}_{S,\epsilon}\Big[\sup_{\mathbf{w}\in\mathcal{W}} \sum_{i=1}^{n}\epsilon_i f(\mathbf{w};\mathbf{z}_i)\Big].$$

Rademacher Complexity

Definition

- Let $\epsilon_1, \ldots, \epsilon_n$ be independent Rademacher variables (taking only values ± 1, with equal probability)
- The Rademacher complexity of a function space $\mathcal{F}$ is defined as ($S = \{z_i\}$)

$$\mathfrak{R}_S(\mathcal{F}) := \mathbb{E}_\epsilon \Big[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^{n} \epsilon_i f(\mathbf{z}_i) \Big]. \tag{8}$$

- We project $\mathcal{F}$ onto S and get a class of vectors in $\mathbb{R}^n$

$$f \mapsto \big(f(\mathbf{z}_1), f(\mathbf{z}_2), \ldots, f(\mathbf{z}_n)\big) \in \mathbb{R}^n.$$

- **Correlation** of f to Rademacher variables:

$$\sum_{i=1}^{n} \epsilon_i f(\mathbf{z}_i) = \big\langle \underbrace{(\epsilon_1, \ldots, \epsilon_n)}_{\text{noise}}, \underbrace{(f(\mathbf{z}_1), \ldots, f(\mathbf{z}_n))}_{\text{projection to } S} \big\rangle$$

- Maximization over $\mathcal{F}$ means the ability of $\mathcal{F}$ to correlate random noise

Rademacher Complexity

We just showed

$$\mathbb{E}\Big[\sup_{\mathbf{w}\in\mathcal{W}}\frac{1}{n}\sum_{i=1}^{n}\Big(\mathbb{E}_{\mathbf{z}}[f(\mathbf{w};\mathbf{z})]-f(\mathbf{w};\mathbf{z}_i)\Big)\Big]\leq\frac{2}{n}\mathbb{E}_{S,\epsilon}\Big[\sup_{\mathbf{w}\in\mathcal{W}}\sum_{i=1}^{n}\epsilon_i f(\mathbf{w};\mathbf{z}_i)\Big]:=2\mathbb{E}_S\mathfrak{R}_S(\mathcal{F}),$$

where $\mathcal{F}$ is the loss function class

$$\mathcal{F}=\big\{\mathbf{z}\mapsto f(\mathbf{w};\mathbf{z}):\mathbf{w}\in\mathcal{W}\big\}.$$

We plug this bound back into Eq. (7), and derive the following theorem.

Thm: Excess risk bound

Let A be the ERM. Then with probability at least $1-\delta$

$$F(A(S))-F(\mathbf{w}^*)\leq\sqrt{2}n^{-\frac{1}{2}}\log^{\frac{1}{2}}(2/\delta)+2\mathbb{E}_S\big[\mathfrak{R}_S(\mathcal{F})\big].$$

- Actually, one can show that $\mathfrak{R}_S(\mathcal{F})$ satisfies the bounded difference condition
- With probability at least $1-\delta/3$, $\mathbb{E}_S\big[\mathfrak{R}_S(\mathcal{F})\big]\leq\mathfrak{R}_S(\mathcal{F})+2^{-1/2}n^{-\frac{1}{2}}\log^{\frac{1}{2}}(3/\delta)$
- Then, with probability at least $1-\delta$ we have the following data-dependent bound

$$F(A(S))-F(\mathbf{w}^*)\leq\frac{2\sqrt{2}\log^{\frac{1}{2}}(3/\delta)}{\sqrt{n}}+2\mathfrak{R}_S(\mathcal{F}).\tag{9}$$

Exact Calculation of Rademacher Complexity

Let $S = \{\mathbf{x}_1, \ldots, \mathbf{x}_d\}$, where $\mathbf{x}_i$ is the i-th unit vector in $\mathbb{R}^d$, i.e., only the i-th component is 1 and other components are 0. Consider the function class

$$\mathcal{F} = \left\{ \mathbf{x} \mapsto \mathbf{w}^\top \mathbf{x} : \mathbf{w} \in \mathbb{R}^d, w^{(j)} \in \{-1, 1\}, j \in [d] \right\}.$$

Then, by the definition we know

$$\begin{aligned}
\mathfrak{R}_S(\mathcal{F}) &= \frac{1}{d} \mathbb{E}_\epsilon \sup_{\mathbf{w} \in \mathbb{R}^d : w^{(j)} \in \{-1,1\}} \sum_{i=1}^{d} \epsilon_i \mathbf{x}_i^\top \mathbf{w} = \frac{1}{d} \mathbb{E}_\epsilon \sup_{\mathbf{w} \in \mathbb{R}^d : w^{(j)} \in \{-1,1\}} \mathbf{w}^\top \sum_{i=1}^{d} \epsilon_i \mathbf{x}_i \\
&= \frac{1}{d} \mathbb{E}_\epsilon \sup_{\mathbf{w} \in \mathbb{R}^d : w^{(j)} \in \{-1,1\}} \mathbf{w}^\top \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_d \end{pmatrix} = \frac{1}{d} \mathbb{E}_\epsilon \sup_{\mathbf{w} \in \mathbb{R}^d : w^{(j)} \in \{-1,1\}} \sum_{j=1}^{d} w^{(j)} \epsilon_j \\
&= \frac{1}{d} \sum_{j=1}^{d} 1 = 1.
\end{aligned}$$

where we have used the observation that $\max_{w^{(j)} \in \{-1,1\}} w^{(j)} \epsilon_j = |\epsilon_j| = 1$.

How to Estimate Rademacher Complexity?

- We reduce the estimation of excess risk to that of Rademacher complexity.
- It suffices to estimate the Rademacher complexity.
- Monte Carlo approach
  - Recall $\mathfrak{R}_S(\mathcal{F}) := \mathbb{E}_\epsilon \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \epsilon_i f(\mathbf{z}_i) \right]$.
  - we randomly draw $(\epsilon_{j1}, \ldots, \epsilon_{jn}), j = 1, \ldots, t$.
  - For each j, we solve the maximization problem

$$\xi_j := \max_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \epsilon_{ji} f(\mathbf{z}_i).$$

  - If t is large, say 100, then $\frac{1}{t} \sum_{j=1}^t \xi_j$ is a good approximation of $\mathfrak{R}_S(\mathcal{F})$
  - However, this is expensive as we need to solve several maximization problems.

Can we directly estimate Rademacher complexity without solving the maximization problem?

Property of Rademacher Complexity

Translation invariance

Consider a function class $\mathcal{F}$ and its translation $\mathcal{F}' = \{f'(\mathbf{z}) = f(\mathbf{z}) + c_0 : f \in \mathcal{F}\}$ for some $c_0 \in \mathbb{R}$. Then $\mathfrak{R}_S(\mathcal{F}) = \mathfrak{R}_S(\mathcal{F}')$.

$$\begin{aligned}\mathfrak{R}_S(\mathcal{F}') &= \mathbb{E}_\epsilon\Big[\sup_{f'\in\mathcal{F}'}\frac{1}{n}\sum_{i=1}^n \epsilon_i f'(\mathbf{z}_i)\Big] = \mathbb{E}_\epsilon\Big[\sup_{f\in\mathcal{F}}\frac{1}{n}\sum_{i=1}^n \epsilon_i (f(\mathbf{z}_i) + c_0)\Big] \\ &= \mathbb{E}_\epsilon\Big[\frac{1}{n}\sum_{i=1}^n \epsilon_i c_0 + \sup_{f\in\mathcal{F}}\frac{1}{n}\sum_{i=1}^n \epsilon_i f(\mathbf{z}_i)\Big] = \mathbb{E}_\epsilon\Big[\sup_{f\in\mathcal{F}}\frac{1}{n}\sum_{i=1}^n \epsilon_i f(\mathbf{z}_i)\Big].\end{aligned}$$

Scaling property

Consider a function class $\mathcal{F}$ and its scaling $\mathcal{F}' = \{f'(\mathbf{z}) = c \cdot f(\mathbf{z}) : f \in \mathcal{F}\}$ for some $c \in \mathbb{R}$. Then $\mathfrak{R}_S(\mathcal{F}') = |c|\mathfrak{R}_S(\mathcal{F})$.

$$\mathfrak{R}_S(\mathcal{F}') = \mathbb{E}_\epsilon\Big[\sup_{f'\in\mathcal{F}'}\frac{1}{n}\sum_{i=1}^n \epsilon_i f'(\mathbf{z}_i)\Big] = \mathbb{E}_\epsilon\Big[\sup_{f\in\mathcal{F}}\frac{1}{n}\sum_{i=1}^n \epsilon_i (cf(\mathbf{z}_i))\Big] = |c|\mathbb{E}_\epsilon\Big[\sup_{f\in\mathcal{F}}\frac{1}{n}\sum_{i=1}^n \epsilon_i f(\mathbf{z}_i)\Big],$$

where we have used ϵ_i and $-\epsilon_i$ have the same distribution.

Property of Rademacher Complexity

We have

$$\mathbb{E}_\epsilon \sup_{f\in\mathcal{F}} \Big| \sum_{i=1}^n \epsilon_i f(\mathbf{z}_i) \Big| \le 2\mathbb{E}_\epsilon \sup_{f\in\{0\}\cup\mathcal{F}} \sum_{i=1}^n \epsilon_i f(\mathbf{z}_i).$$

Note that $|a| = \max\{a, 0\} + \max\{-a, 0\}$ for any a. Then

$$\begin{aligned}
&\mathbb{E}_\epsilon \sup_{f\in\mathcal{F}} \Big| \sum_{i=1}^n \epsilon_i f(\mathbf{z}_i) \Big| \\
&\le \mathbb{E}_\epsilon \sup_{f\in\mathcal{F}} \max\Big\{ \sum_{i=1}^n \epsilon_i f(\mathbf{z}_i), 0 \Big\} + \mathbb{E}_\epsilon \sup_{f\in\mathcal{F}} \max\Big\{ -\sum_{i=1}^n \epsilon_i f(\mathbf{z}_i), 0 \Big\} \\
&= \mathbb{E}_\epsilon \max\Big\{ \sup_{f\in\mathcal{F}} \sum_{i=1}^n \epsilon_i f(\mathbf{z}_i), 0 \Big\} + \mathbb{E}_\epsilon \max\Big\{ \sup_{f\in\mathcal{F}} -\sum_{i=1}^n \epsilon_i f(\mathbf{z}_i), 0 \Big\} \\
&= \mathbb{E}_\epsilon \sup_{f\in\{0\}\cup\mathcal{F}} \sum_{i=1}^n \epsilon_i f(\mathbf{z}_i) + \mathbb{E}_\epsilon \sup_{f\in\{0\}\cup\mathcal{F}} -\sum_{i=1}^n \epsilon_i f(\mathbf{z}_i) \\
&= 2\mathbb{E}_\epsilon \sup_{f\in\{0\}\cup\mathcal{F}} \sum_{i=1}^n \epsilon_i f(\mathbf{z}_i).
\end{aligned}$$

Rademacher Complexity of Convex Hull

Define $\triangle_m = \left\{\mathbf{c} \in \mathbb{R}^m : c_j \geq 0, \sum_{j=1}^m c_j = 1\right\}$.

Convex Hull

The convex hull of $\mathcal{F}$ ($\mathrm{conv}(\mathcal{F})$) is defined as

$$\mathrm{conv}(\mathcal{F}) = \left\{ \sum_{j=1}^m c_j f_j : m \in \mathbb{N}, \mathbf{c} \in \triangle_m, f_j \in \mathcal{F} \right\}.$$

Rademacher complexity of convex hull

We have $\mathfrak{R}_S(\mathrm{conv}(\mathcal{F})) = \mathfrak{R}_S(\mathcal{F})$.

Note for any $\alpha \in \mathbb{R}^m$, $\sup_{\mathbf{c}\in\triangle_m} \sum_{j=1}^m c_j \alpha_j = \max_j \alpha_j$. Then

$$\mathbb{E}_\epsilon \sup_{f\in\mathrm{conv}(\mathcal{F})} \sum_{i=1}^n \epsilon_i f(\mathbf{z}_i) = \mathbb{E}_\epsilon \sup_{\mathbf{c}\in\triangle_m, f_j\in\mathcal{F}} \sum_{i=1}^n \sum_{j=1}^m \epsilon_i c_j f_j(\mathbf{z}_i) = \mathbb{E}_\epsilon \sup_{\mathbf{c}\in\triangle_m, f_j\in\mathcal{F}} \sum_{j=1}^m c_j \sum_{i=1}^n \epsilon_i f_j(\mathbf{z}_i)$$

$$= \mathbb{E}_\epsilon \sup_{\mathbf{c}\in\triangle_m, f_j\in\mathcal{F}} \max_{j\in[m]} \sum_{i=1}^n \epsilon_i f_j(\mathbf{z}_i) = \mathbb{E}_\epsilon \sup_{f\in\mathcal{F}} \sum_{i=1}^n \epsilon_i f(\mathbf{z}_i).$$

Talagrand's Contraction Lemma

Talagrand's Contraction Lemma

Let $\phi_i : \mathbb{R} \mapsto \mathbb{R}$ be G-Lipschitz, i.e., $|\phi_i(a) - \phi_i(b)| \leq G|a - b|$. Then

$$\mathbb{E}_\epsilon \sup_{f \in \mathcal{F}} \sum_{i=1}^{n} \epsilon_i \phi_i(f(\mathbf{z}_i)) \leq G \mathbb{E}_\epsilon \sup_{f \in \mathcal{F}} \sum_{i=1}^{n} \epsilon_i f(\mathbf{z}_i).$$



Michel Talagrand awarded the 2024 Abel Prize

«for his groundbreaking contributions to probability theory and functional analysis, with outstanding applications in mathematical physics and statistics.»

- The contraction lemma is extremely useful to estimate Rademacher complexity
- It removes the nonlinear function ϕ

$$\mathfrak{R}_S(\phi \circ \mathcal{F}) \leq G\mathfrak{R}_S(\mathcal{F}),$$

where $\phi \circ \mathcal{F} = \{\mathbf{z} \mapsto \phi(f(\mathbf{z})) : f \in \mathcal{F}\}$ is the composition function class.

Proof of Talagrand's Contraction Lemma

$$\mathbb{E}_{\epsilon_{1:n}}\Big[\sup_{f\in\mathcal{F}}\sum_{i=1}^{n}\epsilon_i\phi_i(f(\mathbf{z}_i))\Big]$$

$$= \mathbb{E}_{\epsilon_{1:n-1}}\frac{1}{2}\Big\{\sup_{f\in\mathcal{F}}\Big(\sum_{i=1}^{n-1}\epsilon_i\phi_i(f(\mathbf{z}_i))+\phi_n(f(\mathbf{z}_n))\Big)+\sup_{f'\in\mathcal{F}}\Big(\sum_{i=1}^{n-1}\epsilon_i\phi_i(f'(\mathbf{z}_i))-\phi_n(f'(\mathbf{z}_n))\Big)\Big\}$$

$$= \mathbb{E}_{\epsilon_{1:n-1}}\frac{1}{2}\Big\{\sup_{f,f'\in\mathcal{F}}\Big(\sum_{i=1}^{n-1}\big(\epsilon_i\phi_i(f(\mathbf{z}_i))+\epsilon_i\phi_i(f'(\mathbf{z}_i))\big)+\phi_n(f(\mathbf{z}_n))-\phi_n f'(\mathbf{z}_n)\Big)\Big\}$$

$$\leq \mathbb{E}_{\epsilon_{1:n-1}}\frac{1}{2}\Big\{\sup_{f,f'\in\mathcal{F}}\Big(\sum_{i=1}^{n-1}\big(\epsilon_i\phi_i(f(\mathbf{z}_i))+\epsilon_i\phi_i(f'(\mathbf{z}_i))\big)+G|f(\mathbf{z}_n)-f'(\mathbf{z}_n)|\Big)\Big\}$$

$$= \mathbb{E}_{\epsilon_{1:n-1}}\frac{1}{2}\Big\{\sup_{f,f'\in\mathcal{F}}\Big(\sum_{i=1}^{n-1}\big(\epsilon_i\phi_i(f(\mathbf{z}_i))+\epsilon_i\phi_i(f'(\mathbf{z}_i))\big)+G\big(f(\mathbf{z}_n)-f'(\mathbf{z}_n)\big)\Big)\Big\}$$

$$= \mathbb{E}_{\epsilon_{1:n-1}}\frac{1}{2}\Big\{\sup_{f\in\mathcal{F}}\Big(\sum_{i=1}^{n-1}\epsilon_i\phi_i(f(\mathbf{z}_i))+Gf(\mathbf{z}_n)\Big)+\sup_{f'\in\mathcal{F}}\Big(\sum_{i=1}^{n-1}\epsilon_i\phi_i(f'(\mathbf{z}_i))-Gf'(\mathbf{z}_n)\Big)\Big\}$$

$$= \mathbb{E}_{\epsilon_{1:n}}\sup_{f\in\mathcal{F}}\Big(\sum_{i=1}^{n-1}\epsilon_i\phi_i(f(\mathbf{z}_i))+\epsilon_n Gf(\mathbf{z}_n)\Big).$$

We repeat this process n times and get the stated bound!

Excess Risk Bounds for Lipschitz Loss

- We just derived the following data-dependent bound (Eq. (9))

$$F(A(S)) - F(\mathbf{w}^*) \leq \frac{2\sqrt{2}\log^{\frac{1}{2}}(3/\delta)}{\sqrt{n}} + 2\mathfrak{R}_S(\mathcal{F}).$$

- Suppose ℓ is G-Lipschitz, i.e., $|\ell(a,y) - \ell(a',y)| \leq G|a-a'|$. Then Talagrand's contraction lemma implies

$$\mathfrak{R}_S(\mathcal{F}) = \mathfrak{R}_S\Big(\mathbf{z} \mapsto \ell(h_{\mathbf{w}}(\mathbf{x}), y) : \mathbf{w} \in \mathcal{W}\Big)$$

$$= \frac{1}{n}\mathbb{E}_\epsilon \sup_{\mathbf{w}\in\mathcal{W}} \Big[\sum_{i=1}^n \ell(h_{\mathbf{w}}(\mathbf{x}_i), y_i)\Big] \leq \frac{G}{n}\mathbb{E}_\epsilon \sup_{\mathbf{w}\in\mathcal{W}} \Big[\sum_{i=1}^n h_{\mathbf{w}}(\mathbf{x}_i)\Big] = G\mathfrak{R}_S(\mathcal{H}),$$

where $\mathcal{H} = \{\mathbf{x} \mapsto h_{\mathbf{w}}(\mathbf{x}) : \mathbf{w} \in \mathcal{W}\}$, $\mathcal{F} = \{\mathbf{z} \mapsto \ell(h_{\mathbf{w}}(\mathbf{x}), y) : \mathbf{w} \in \mathcal{W}\}$.

With Lipschitzness, we reduce the Rademacher complexity of the loss function class to that of the hypothesis space!

Excess risk bounds for Lipschitz Loss

Suppose ℓ is G-Lipschitz. Then with probability at least $1-\delta$

$$F(A(S)) - F(\mathbf{w}^*) \leq \frac{2\sqrt{2}\log^{\frac{1}{2}}(3/\delta)}{\sqrt{n}} + 2G\mathfrak{R}_S(\mathcal{H}). \tag{10}$$

Excess risk bounds for Lipschitz Loss

- We just reduced the Rademacher complexity of $\mathcal{F}$ to that of $\mathcal{H}$ for Lipschitz loss. Examples include
  - logistic loss: $\ell(\hat{y}, y) = \log(1 + \exp(-y\hat{y}))$
  - hinge loss: $\ell(\hat{y}, y) = \max\{0, 1 - y\hat{y}\}$
- $\mathfrak{R}_S(\mathcal{H})$ is much more convenient to handle
- We will give examples on the estimation of Rademacher complexity
  - finite function class
  - linear model
  - shallow neural networks

Rademacher Complexity for a Finite Class of Functions

Massart's Lemma. Let A be a finite set of vectors in $\mathbb{R}^n$ with $\|\mathbf{a}\|_2 \le r$ for any $\mathbf{a} \in A$.

$$\mathbb{E}\Big[\sup_{\mathbf{a}\in A} \frac{1}{n}\sum_{i=1}^{n} \epsilon_i a_i\Big] \le \frac{r\big(2\log|A|\big)^{\frac{1}{2}}}{n}.$$

Proof. Note $\exp\big(\mathbb{E}[X]\big) \le \mathbb{E}[\exp(X)]$ (Jensen's inequality). Then, for any $\lambda > 0$

$$\begin{aligned}
\exp\Big(\lambda\mathbb{E}\Big[\sup_{\mathbf{a}\in A}\sum_{i=1}^{n}\epsilon_i a_i\Big]\Big) &\le \mathbb{E}\Big[\exp\Big(\lambda\sup_{\mathbf{a}\in A}\sum_{i=1}^{n}\epsilon_i a_i\Big)\Big] = \mathbb{E}\Big[\sup_{\mathbf{a}\in A}\exp\Big(\lambda\sum_{i=1}^{n}\epsilon_i a_i\Big)\Big] \\
&\le \sum_{\mathbf{a}\in A}\mathbb{E}\Big[\exp\Big(\lambda\sum_{i=1}^{n}\epsilon_i a_i\Big)\Big] = \sum_{\mathbf{a}\in A}\prod_{i=1}^{n}\mathbb{E}\big[\exp(\lambda\epsilon_i a_i)\big] \\
&= \sum_{\mathbf{a}\in A}\prod_{i=1}^{n}\frac{1}{2}\Big(\exp(\lambda a_i)+\exp(-\lambda a_i)\Big) \le \sum_{\mathbf{a}\in A}\prod_{i=1}^{n}\exp\big(\lambda^2 a_i^2/2\big) \\
&= \sum_{\mathbf{a}\in A}\exp\big(\lambda^2\|\mathbf{a}\|_2^2/2\big) \le |A|\exp(\lambda^2 r^2/2),
\end{aligned}$$

where we use $(e^x + e^{-x})/2 \le \exp(x^2/2)$. Taking log and dividing by λ

$$\mathbb{E}\Big[\sup_{\mathbf{a}\in A}\sum_{i=1}^{n}\epsilon_i a_i\Big] \le \frac{\log|A|}{\lambda} + \frac{\lambda r^2}{2} \quad (\lambda = r^{-1}\sqrt{2\log|A|}).$$

Rademacher Complexity for a Finite Class of Functions

Finite Class

Let $\mathcal{F}$ be a finite set of functions such that $|f(\mathbf{z})| \leq 1$. Then,

$$\mathfrak{R}_S(\mathcal{F}) \leq \left(\frac{2\log|\mathcal{F}|}{n}\right)^{\frac{1}{2}}.$$

- Consider the set of vectors $A = \left\{\left(f(\mathbf{z}_1), \ldots, f(\mathbf{z}_n)\right) : f \in \mathcal{F}\right\}$.
- Then for every $\mathbf{a} \in A$, we have $\|\mathbf{a}\| \leq \sqrt{n}$.
- Applying Massart's Lemma shows

$$\mathfrak{R}_S(\mathcal{F}) = \frac{1}{n}\mathbb{E}\left[\sup_{\mathbf{a}\in A}\sum_{i=1}^{n}\epsilon_i a_i\right] \leq \left(\frac{2\log|\mathcal{F}|}{n}\right)^{\frac{1}{2}}.$$

Rademacher Complexity for Linear Function Class

Consider the linear function class $\mathcal{H} = \{\mathbf{x} \mapsto \mathbf{w}^\top \mathbf{x} : \|\mathbf{w}\|_2 \leq 1\}$.

$$\text{Then} \quad \mathfrak{R}_S(\mathcal{H}) = \frac{1}{n}\mathbb{E}_\epsilon \sup_{h\in\mathcal{H}} \sum_{i=1}^n \epsilon_i h(\mathbf{x}_i) = \frac{1}{n}\mathbb{E}_\epsilon \sup_{h\in\mathcal{H}} \sum_{i=1}^n \epsilon_i \mathbf{w}^\top \mathbf{x}_i$$

$$= \frac{1}{n}\mathbb{E}_\epsilon \sup_{\mathbf{w}:\|\mathbf{w}\|_2\leq 1} \mathbf{w}^\top \Big(\sum_{i=1}^n \epsilon_i \mathbf{x}_i\Big) \leq \frac{1}{n}\mathbb{E}_\epsilon \sup_{\mathbf{w}:\|\mathbf{w}\|_2\leq 1} \|\mathbf{w}\|_2 \Big\|\sum_{i=1}^n \epsilon_i \mathbf{x}_i\Big\|_2$$

$$\leq \frac{1}{n}\mathbb{E}_\epsilon \Big(\Big\|\sum_{i=1}^n \epsilon_i \mathbf{x}_i\Big\|_2^2\Big)^{\frac{1}{2}}.$$

Since $\mathbb{E}[\sqrt{f(X)}] \leq \sqrt{\mathbb{E}[f(X)]}$, we further know

$$\mathfrak{R}_S(\mathcal{H}) \leq \frac{1}{n}\Big(\mathbb{E}_\epsilon \Big\|\sum_{i=1}^n \epsilon_i \mathbf{x}_i\Big\|_2^2\Big)^{\frac{1}{2}} = \frac{1}{n}\Big(\mathbb{E}_\epsilon \sum_{i,j=1}^n \epsilon_i\epsilon_j \mathbf{x}_i^\top \mathbf{x}_j\Big)^{\frac{1}{2}}$$

$$= \frac{1}{n}\Big(\mathbb{E}_\epsilon \sum_{i=1}^n \epsilon_i^2 \mathbf{x}_i^\top \mathbf{x}_i + \sum_{i\neq j} \mathbb{E}_\epsilon \epsilon_i\epsilon_j \mathbf{x}_i^\top \mathbf{x}_j\Big)^{\frac{1}{2}} = \frac{1}{n}\Big(\sum_{i=1}^n \|\mathbf{x}_i\|_2^2\Big)^{\frac{1}{2}}.$$

$$\Longrightarrow F(A(S)) - F(\mathbf{w}^*) \leq \frac{2\sqrt{2}\log^{\frac{1}{2}}(3/\delta)}{\sqrt{n}} + \frac{2G}{n}\Big(\sum_{i=1}^n \|\mathbf{x}_i\|_2^2\Big)^{\frac{1}{2}}.$$

Rademacher Complexity for Shallow ReLU Networks

Consider shallow ReLU networks with $\sigma(t) = \max\{t, 0\}$

$$\mathcal{H} = \Big\{ \sum_{j=1}^{m} a_j \sigma(\mathbf{w}_j^\top \mathbf{x}) : \sum_{j=1}^{m} |a_j| \|\mathbf{w}_j\|_2 \le R \Big\}.$$

Note the positive homogeneity $\sigma(ta) = a\sigma(t)$ for any $a \ge 0$:

$$\begin{aligned}
n\mathfrak{R}_S(\mathcal{H}) &= \mathbb{E}_\epsilon \sup_{h\in\mathcal{H}} \sum_{i=1}^{n} \epsilon_i h(\mathbf{x}_i) = \mathbb{E}_\epsilon \sup_{h\in\mathcal{H}} \sum_{i=1}^{n} \epsilon_i \sum_{j=1}^{m} a_j \sigma(\mathbf{w}_j^\top \mathbf{x}_i) \\
&= \mathbb{E}_\epsilon \sup_{h\in\mathcal{H}} \sum_{i=1}^{n} \epsilon_i \sum_{j=1}^{m} a_j \|\mathbf{w}_j\|_2 \sigma\big(\mathbf{w}_j^\top \mathbf{x}_i / \|\mathbf{w}_j\|_2\big) \\
&= \mathbb{E}_\epsilon \sup_{h\in\mathcal{H}} \sum_{j=1}^{m} |a_j| \|\mathbf{w}_j\|_2 \mathrm{sgn}(a_j) \sum_{i=1}^{n} \epsilon_i \sigma\big(\mathbf{w}_j^\top \mathbf{x}_i / \|\mathbf{w}_j\|_2\big) \\
&\le \mathbb{E}_\epsilon \sup_{h\in\mathcal{H}} \sum_{j=1}^{m} |a_j| \|\mathbf{w}_j\|_2 \max_{j\in[m]} \Big| \mathrm{sgn}(a_j) \sum_{i=1}^{n} \epsilon_i \sigma\big(\mathbf{w}_j^\top \mathbf{x}_i / \|\mathbf{w}_j\|_2\big) \Big| \\
&\le R\mathbb{E}_\epsilon \sup_{\mathbf{w}} \Big| \sum_{i=1}^{n} \epsilon_i \sigma\big(\mathbf{w}^\top \mathbf{x}_i / \|\mathbf{w}\|_2\big) \Big|.
\end{aligned}$$

Rademacher Complexity for Shallow ReLU Networks

We just showed that $n\mathfrak{R}_S(\mathcal{H}) \le R\mathbb{E}_\epsilon \sup_{\mathbf{w}} \left| \sum_{i=1}^n \epsilon_i \sigma\left(\mathbf{w}^\top \mathbf{x}_i / \|\mathbf{w}\|_2\right) \right|$. Then

$$
\begin{aligned}
n\mathfrak{R}_S(\mathcal{H}) &\le R\mathbb{E}_\epsilon \sup_{\tilde{\mathbf{w}}:\|\tilde{\mathbf{w}}\|_2 \le 1} \left| \sum_{i=1}^n \epsilon_i \sigma\left(\tilde{\mathbf{w}}^\top \mathbf{x}_i\right) \right| \\
&\le 2R\mathbb{E}_\epsilon \sup_{\tilde{\mathbf{w}}:\|\tilde{\mathbf{w}}\|_2 \le 1} \sum_{i=1}^n \epsilon_i \sigma\left(\tilde{\mathbf{w}}^\top \mathbf{x}_i\right) \le 2R\mathbb{E}_\epsilon \sup_{\tilde{\mathbf{w}}:\|\tilde{\mathbf{w}}\|_2 \le 1} \sum_{i=1}^n \epsilon_i \tilde{\mathbf{w}}^\top \mathbf{x}_i \\
&\le 2R\mathbb{E}_\epsilon \sup_{\tilde{\mathbf{w}}:\|\tilde{\mathbf{w}}\|_2 \le 1} \tilde{\mathbf{w}}^\top \left(\sum_{i=1}^n \epsilon_i \mathbf{x}_i \right) \le 2R\mathbb{E}_\epsilon \left\| \sum_{i=1}^n \epsilon_i \mathbf{x}_i \right\|_2 \le 2R \left(\sum_{i=1}^n \|\mathbf{x}_i\|_2^2 \right)^{\frac{1}{2}}.
\end{aligned}
$$

For $\mathcal{H} = \left\{ \sum_{j=1}^m a_j \sigma(\mathbf{w}_j^\top \mathbf{x}) : \sum_{j=1}^m |a_j| \|\mathbf{w}_j\|_2 \le R \right\}$ with $\sigma(t) = \max\{t, 0\}$, we know

$$
\mathfrak{R}_S(\mathcal{H}) \le \frac{2R}{n} \left(\sum_{i=1}^n \|\mathbf{x}_i\|_2^2 \right)^{\frac{1}{2}}. \tag{11}
$$

A width-independent Rademacher complexity bound!

Applications

Applications to Support Vector Machines

Generalization of ERM with SVMs

- Consider a linear model
$$\mathbf{x} \mapsto h_{\mathbf{w}}(\mathbf{x}) := \mathbf{w}^\top \mathbf{x}.$$
- Consider ERM on $\mathcal{H} = \left\{\mathbf{x} \mapsto h_{\mathbf{W}}(\mathbf{x}) : \|\mathbf{w}\|_2 \leq 1\right\}$
- Consider the hinge loss $f(\mathbf{w}; \mathbf{z}) = \max\{0, 1 - y h_{\mathbf{w}}(\mathbf{x})\}$
- The loss function class becomes $\mathcal{F} = \left\{\mathbf{z} \mapsto \max\{0, 1 - y h_{\mathbf{w}}(\mathbf{x})\} : h_{\mathbf{w}} \in \mathcal{H}\right\}$.

To apply our general theory, we require to control Rademacher complexity of loss function class and $f(\mathbf{W}; \mathbf{z})$

Generalization of ERM with SVMs

- The map $t \mapsto \max\{0, 1 - yt\}$ is 1-Lipschitz.By Talagrand's lemma, we know $\mathfrak{R}_S(\mathcal{F}) \leq \mathfrak{R}_S(\mathcal{H})$, which can be further bounded by $\frac{1}{n}\left(\sum_{i=1}^n \|\mathbf{x}_i\|_2^2\right)^{\frac{1}{2}}$.
- If $\|\mathbf{x}\|_2 \leq 1$, we know (note $\sigma(t) \leq |t|$)

$$f(\mathbf{w}; \mathbf{z}) = \max\{0, 1 - y\langle \mathbf{w}, \mathbf{x}\rangle\} \leq 1 + |y\langle \mathbf{w}, \mathbf{x}\rangle| \leq 2.$$

- By Eq. (10), with probability at least $1 - \delta$, ERM satisfies

$$F(A(S)) - F(\mathbf{w}^*) \lesssim \frac{\log^{\frac{1}{2}}(1/\delta)}{\sqrt{n}} + \frac{1}{n}\left(\sum_{i=1}^n \|\mathbf{x}_i\|_2^2\right)^{\frac{1}{2}}.$$

Applications to Shallow Neural Networks

Generalization of ERM with SNNs ($\sigma(t) = \max\{t, 0\}$)

- Consider an SNN

$$\mathbf{x} \mapsto h_{\mathbf{W}}(\mathbf{x}) := \sum_{j=1}^{m} a_j \sigma(\mathbf{w}_j^\top \mathbf{x}), \quad \mathbf{W} = (\mathbf{w}_1, \ldots, \mathbf{w}_m).$$

- Consider ERM on $\mathcal{H} = \left\{\mathbf{x} \mapsto h_{\mathbf{W}}(\mathbf{x}) : \sum_{j=1}^{m} |a_j| \|\mathbf{w}_j\|_2 \leq 1\right\}$
- Consider the hinge loss $f(\mathbf{W}; \mathbf{z}) = \max\{0, 1 - y h_{\mathbf{W}}(\mathbf{x})\}$
- The loss function class becomes $\mathcal{F} = \left\{\mathbf{z} \mapsto \max\{0, 1 - y h_{\mathbf{W}}(\mathbf{x})\} : h_{\mathbf{W}} \in \mathcal{H}\right\}$.

To apply our general theory, we require to control Rademacher complexity of loss function class and $f(\mathbf{W}; \mathbf{z})$

Generalization of ERM with SNNs ($\sigma(t) = \max\{t, 0\}$)

- The map $t \mapsto \max\{0, 1 - yt\}$ is 1-Lipschitz.By Talagrand's lemma, we know $\mathfrak{R}_S(\mathcal{F}) \leq \mathfrak{R}_S(\mathcal{H})$, which can be further bounded by $\frac{2}{n}\left(\sum_{i=1}^{n} \|\mathbf{x}_i\|_2^2\right)^{\frac{1}{2}}$.
- If $\|\mathbf{x}\|_2 \leq 1$, we know (note $\sigma(t) \leq |t|$)

$$|h_{\mathbf{W}}(\mathbf{x})| = \left|\sum_{j=1}^{m} a_j \sigma(\mathbf{w}_j^\top \mathbf{x})\right| \leq \sum_{j=1}^{m} |a_j||\mathbf{w}_j^\top \mathbf{x}| \leq \|\mathbf{x}\|_2 \sum_{j=1}^{m} |a_j| \|\mathbf{w}_j\|_2 \leq 1.$$

- Then $f(\mathbf{W}; \mathbf{z}) \leq 2$. By Eq. (10), with probability at least $1 - \delta$, ERM satisfies

$$F(A(S)) - F(\mathbf{w}^*) \lesssim \frac{\log^{\frac{1}{2}}(1/\delta)}{\sqrt{n}} + \frac{1}{n}\left(\sum_{i=1}^{n} \|\mathbf{x}_i\|_2^2\right)^{\frac{1}{2}}.$$

Width-independent Generalization Bounds!

Growth Function and VC Dimension for Binary Classification

Growth Function

- Massart's lemma gives a Rademacher complexity estimate for a finite function class
- However, the hypothesis space is often very large and contains an infinite number of hypothesis
- What matters is the projection of the function space onto training dataset S
  - For binary classification, projection of $h \in \mathcal{H}$ onto S is an n-dimensional vector
  - Each component is either 1 or -1
  - Therefore, the cardinality of $\mathcal{H}_S := \{(h(\mathbf{x}_1), \ldots, h(\mathbf{x}_n))\}$is at most 2^n
  - If the cardinality is 2^n, then Massart's lemma implies

$$\mathfrak{R}_S(\mathcal{H}) \leq \left(\frac{2 \log |\mathcal{H}_S|}{n}\right)^{\frac{1}{2}} = \left(\frac{2n}{n}\right)^{\frac{1}{2}} = \sqrt{2}$$

  - This leads to a vacuous bound

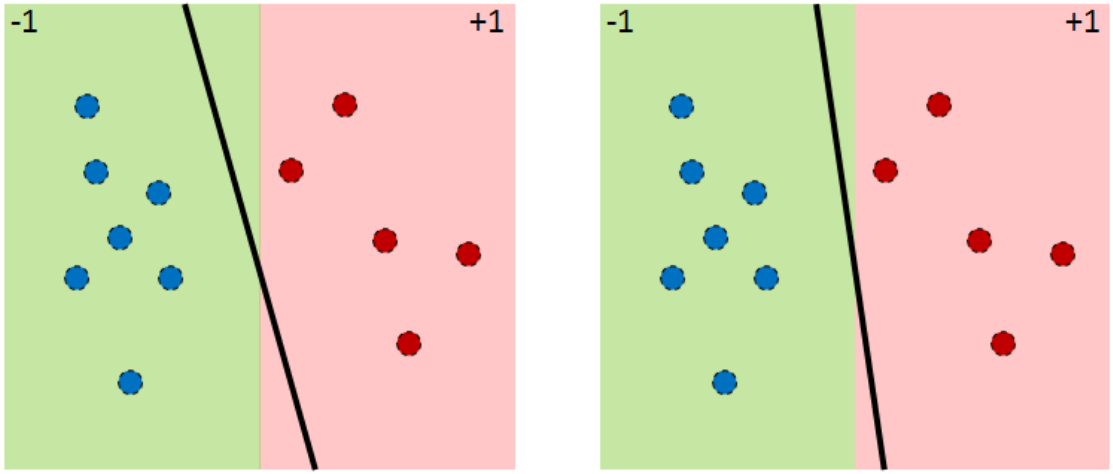
Fortunately, $\mathcal{H}_S$ is often much smaller!

Dichotomies

- Dichotomy = mini-hypothesis

Hypothesis	Dichotomy
$h : \mathcal{X} \mapsto \{+1, -1\}$	$h : \{\mathbf{x}_1, \ldots, \mathbf{x}_n\} \mapsto \{+1, -1\}$
for all population samples	for training samples only
number can be infinite	number is at most 2^n

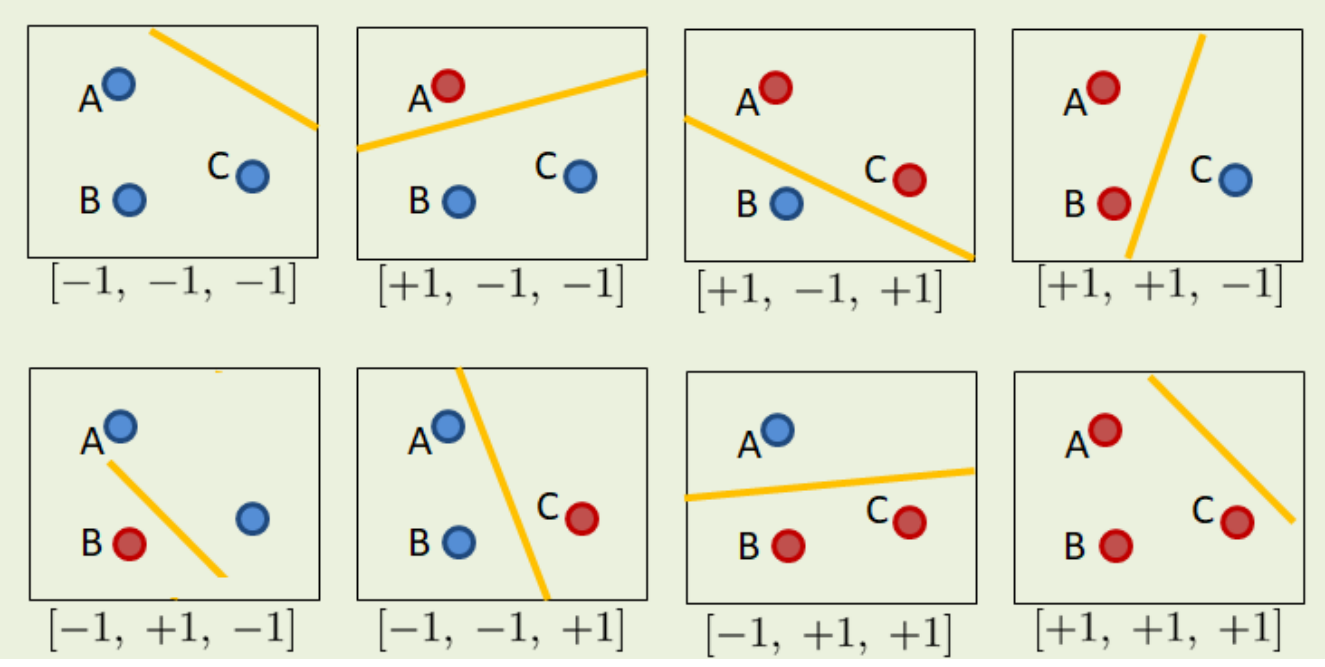
- Different hypothesis, the same dichotomy



Dichotomies

Let $S = \{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$. The dichotomies generated by $\mathcal{H}$ on these points are

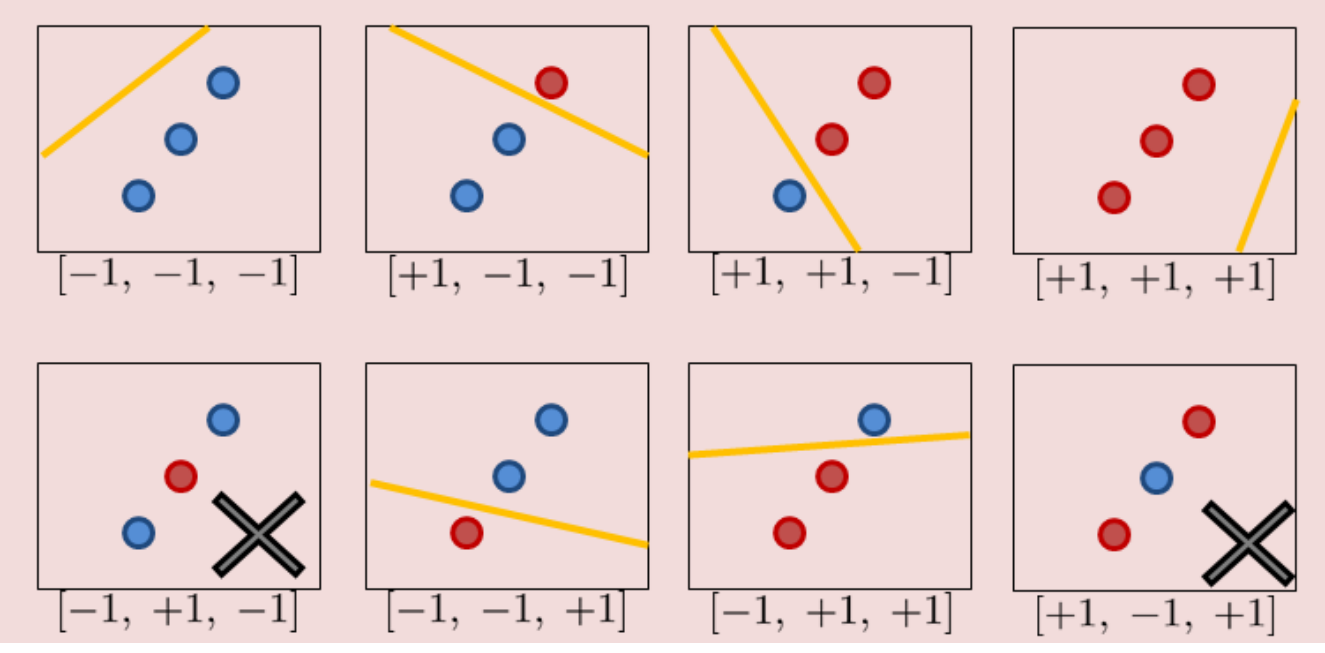
$$\mathcal{H}_S = \{(h(\mathbf{x}_1), \ldots, h(\mathbf{x}_n)) : h \in \mathcal{H}\}$$



Dichotomies

Let $S = \{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$. The dichotomies generated by $\mathcal{H}$ on these points are

$$\mathcal{H}_S = \{(h(\mathbf{x}_1), \ldots, h(\mathbf{x}_n)) : h \in \mathcal{H}\}$$



Growth Function

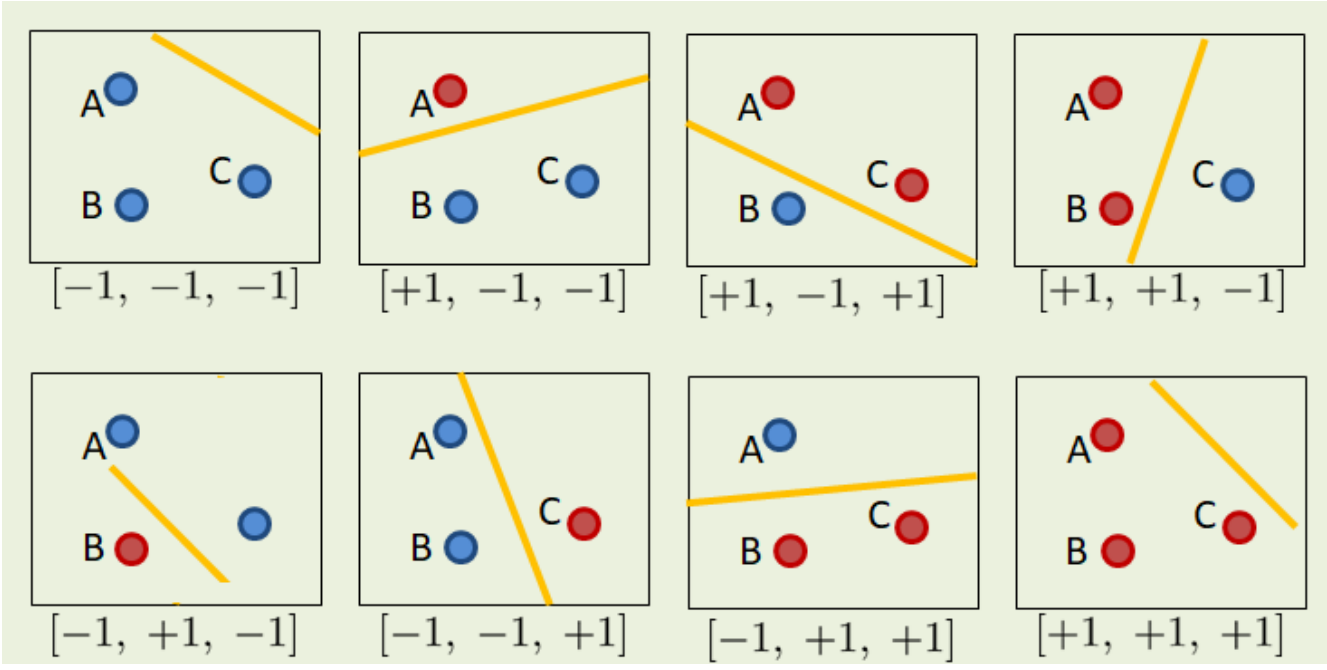
The growth function is defined as

$$m_{\mathcal{H}}(n) = \max_{S:|S|=n} \left|\mathcal{H}_S\right|.$$

- Intuitively, the growth function is the largest cardinality of the projection of $\mathcal{H}$ on any samples of cardinality n
- $m_{\mathcal{H}}(n)$ represents the expressiveness of $\mathcal{H}$
- It only depends on $\mathcal{H}$ and n, does not depend on the learning algorithm and S

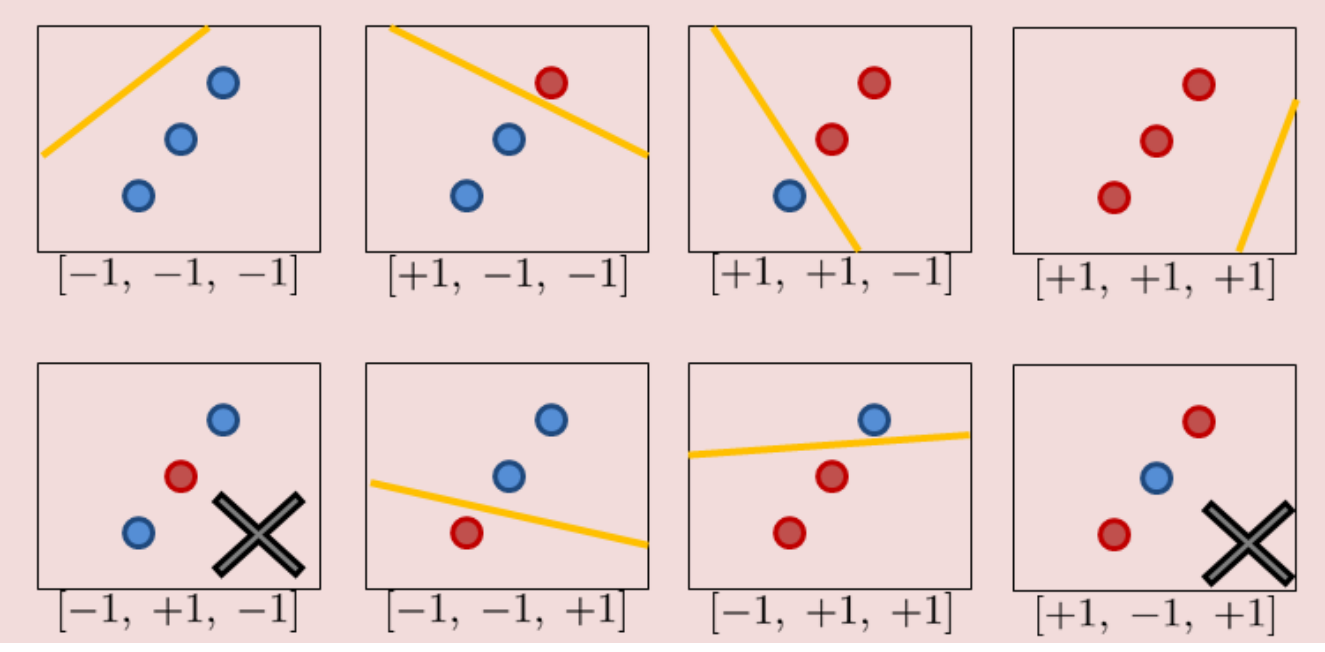
Examples of $m_{\mathcal{H}}(n)$

- $\mathcal{H} =$ linear models in $2D$
- $n = 3$
- How many dichotomies can I generate by moving the three points
- This gives you 8. Are we the best?

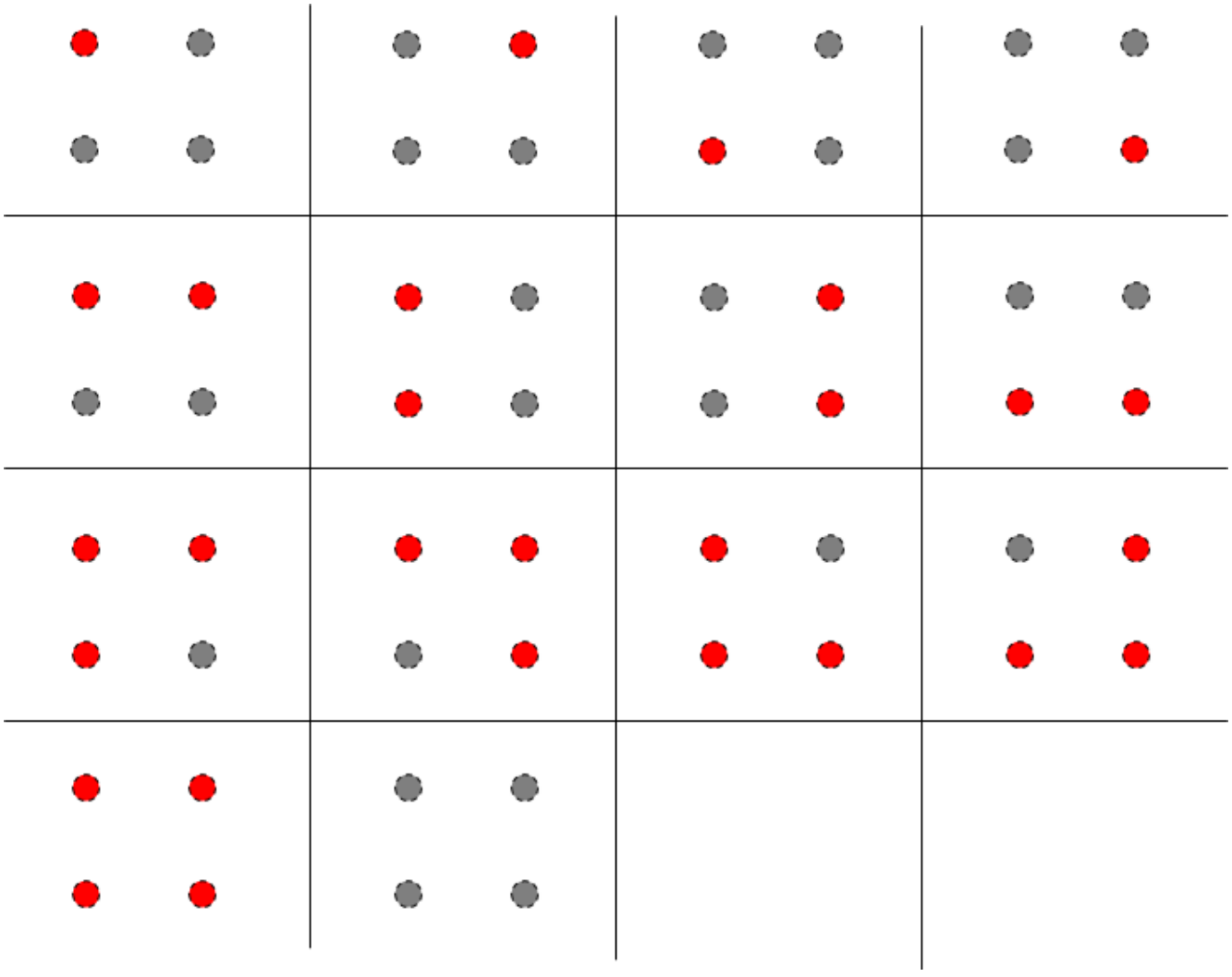


Examples of $m_{\mathcal{H}}(n)$

- $\mathcal{H} =$ linear models in $2D$
- $n = 3$
- How many dichotomies can I generate by moving the three points
- This gives you 6. The previous is the best. So $m_{\mathcal{H}}(3) = 8$



What about $m_{\mathcal{H}}(4)$ for linear $\mathcal{H}$ in 2D? Ans: 14

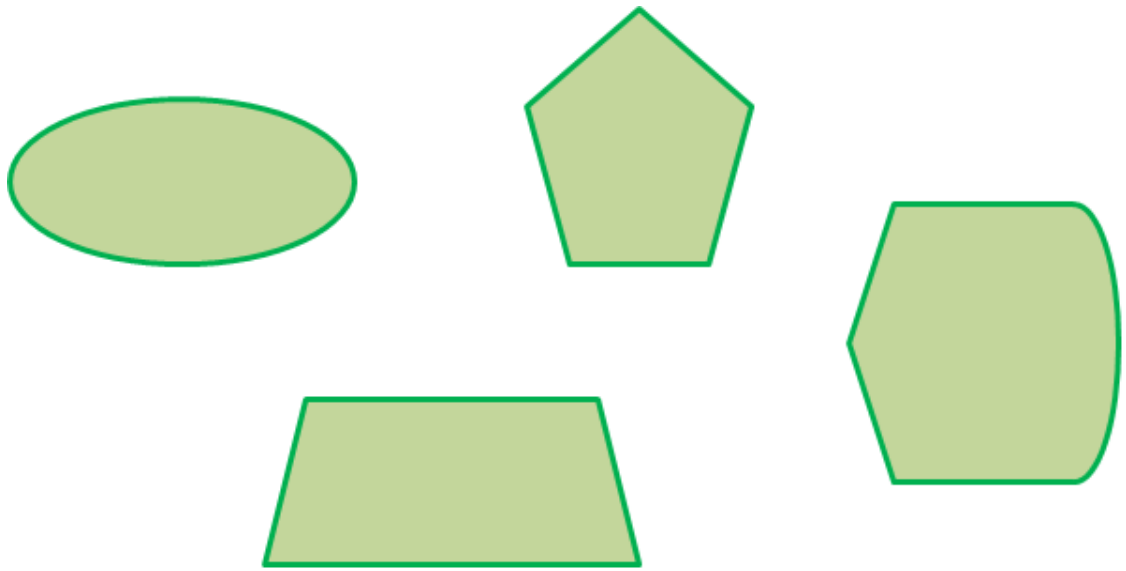


Another Example

Let

$$\mathcal{H} = \left\{ h : \mathbb{R}^2 \mapsto \{+1, -1\} \text{ s.t. } \{\mathbf{x} : h(\mathbf{x}) = 1\} \text{ is convex} \right\}.$$

That is, $h \in \mathcal{H}$ iff $h^{-1}(1)$ is a convex subset of $\mathbb{R}^2$.



Another Example

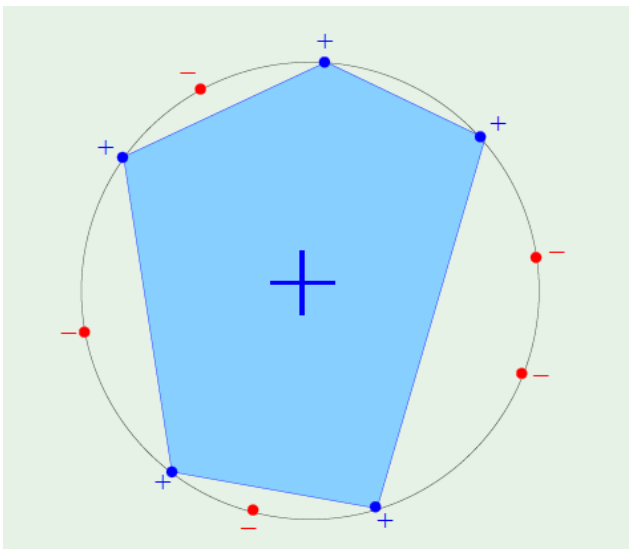
- Put all n points on a circle.
- Then for each labeling of these n examples
  - Let $\mathbf{x}_{i_1}, \ldots, \mathbf{x}_{i_m}$ be positive examples
  - Then you construct the convex hull of these positive examples

$$\mathcal{X}_+ = \Big\{ \sum_{j=1}^{m} \alpha_j \mathbf{x}_{i_j} : \alpha_j \in [0,1], \sum_{j=1}^{m} \alpha_{i_j} = 1 \Big\}.$$

Then you define

$$h(\mathbf{x}) = \begin{cases} 1, & \text{if } \mathbf{x} \in \mathcal{X}_+ \\ -1, & \text{otherwise.} \end{cases}$$

Then $h(\mathbf{x}_i) = -1$ if and only if $\mathbf{x}_i$ is a negative example



We can get 2^n different dichotomies, and this is the best possible one

$$m_{\mathcal{H}}(n) = 2^n, \quad \forall n \in \mathbb{N}.$$

Shatter and VC Dimension

If a hypothesis set $\mathcal{H}$ is able to generate all 2^n dichotomies, then we say $\mathcal{H}$ shatters $\mathbf{x}_1, \ldots, \mathbf{x}_n$.

VC dimension

The Vapnik-Chervonenkis dimension of a hypothesis set $\mathcal{H}$, denoted by d_{VC}, is the largest value of n for which $\mathcal{H}$ can shatter all n training samples

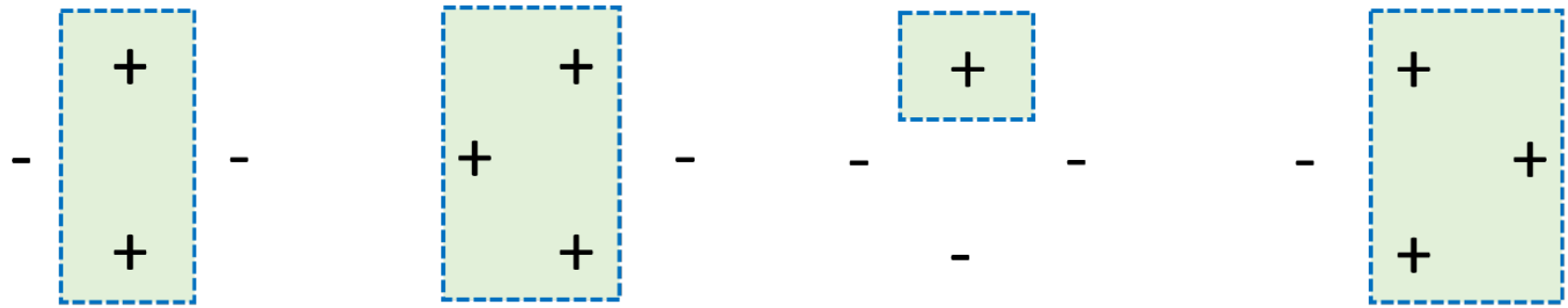
To show that $d_{\text{VC}}(\mathcal{H}) = d$ we need to show

- $d_{\text{VC}}(\mathcal{H}) \geq d$: there exists a set S of size n that can be shattered by $\mathcal{H}$
- $d_{\text{VC}}(\mathcal{H}) < d + 1$: every set of size $d + 1$ cannot be shattered by $\mathcal{H}$

VC Dimension: Example

What is the VC dimension of a $2D$ classifier with a rectangle shape?

- You can try putting 4 data points in whatever way
- There will be 16 possible configurations
- You can show that the rectangle classifier can shatter all these 16 points



- If you do 5 datapoints, then not possible (put one negative in the interior, and four positive at the boundary)
- So VC dimension is 4

Example: Linear Classifiers

Example (Linear classifiers)

Let $\operatorname{sgn}(a)$ denote the sign of a. Consider

$$\mathcal{H} = \{h(\mathbf{x}) = \operatorname{sgn}\Big(\sum_{j=1}^{d} w^{(j)} x^{(j)}\Big) : \mathbf{w} \in \mathbb{R}^d\}.$$

Then $d_{\mathrm{VC}}(\mathcal{H}) = d$.

It suffices to show $d_{\mathrm{VC}}(\mathcal{H}) \geq d$ and $d_{\mathrm{VC}}(\mathcal{H}) \leq d$!

Example: Linear Classifiers

To prove $d_{\mathrm{VC}}(\mathcal{H}) \geq d$, we can build d training examples which can be shattered.

Let $\mathbf{x}_j = (\ldots, 0, 1, 0, \ldots)^\top$ (i.e., the j-th unit vector) for $j = 1, \ldots, d$. Then

$$\begin{pmatrix} h(\mathbf{x}_1) \\ h(\mathbf{x}_2) \\ \vdots \\ h(\mathbf{x}_d) \end{pmatrix} = \operatorname{sgn} \begin{pmatrix} \mathbf{x}_1^\top \mathbf{w} \\ \mathbf{x}_2^\top \mathbf{w} \\ \vdots \\ \mathbf{x}_d^\top \mathbf{w} \end{pmatrix} = \operatorname{sgn} \left(\underbrace{\begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & \vdots \\ 0 & \ddots & \ddots & 0 \\ \vdots & \vdots & 0 & 1 \end{pmatrix}}_{:=X} \underbrace{\begin{pmatrix} w^{(1)} \\ w^{(2)} \\ \vdots \\ w^{(d)} \end{pmatrix}}_{:=\mathbf{w}} \right).$$

For any $\mathbf{y} = (y_1, \ldots, y_d)^\top \in \{+1, -1\}^d$, can we find $\mathbf{w}$ such that $\operatorname{sgn}(X\mathbf{w}) = \mathbf{y}$?

It is clear that X is invertible, then we just set $\mathbf{w} = X^{-1}\mathbf{y}$!

Example: Linear Classifiers

- To show $d_{\mathrm{VC}}(\mathcal{H}) \leq d$, we need to show that it cannot shatter any set of $d+1$ points.
- Consider any $d+1$ points $\mathbf{x}_1, \ldots, \mathbf{x}_{d+1}$. There are more points than dimensions, and therefore they are linearly dependent.
- Then, we can find a_i (not all equal to zero) such that

$$\mathbf{x}_j = \sum_{i:i \neq j} a_i \mathbf{x}_i.$$

- Now we construct a dichotomy that cannot be generated: $y_i = \mathrm{sgn}(a_i)$ for $i \neq j$, and $y_j = -1$.
- For any $\mathbf{w}$ we have

$$\mathbf{w}^\top \mathbf{x}_j = \sum_{i:i \neq j} a_i \mathbf{w}^\top \mathbf{x}_i > 0 \Longrightarrow \mathrm{sgn}(\mathbf{w}^\top \mathbf{x}_j) = +1.$$

VC Dimension and Rademacher Complexity

Sauer's Lemma

Let d_{VC} be the VC dimension of a hypothesis set $\mathcal{H}$. Then

$$m_{\mathcal{H}}(n) \leq \sum_{i=0}^{d_{\mathrm{VC}}} \binom{n}{i} \leq n^{d_{\mathrm{VC}}} + 1.$$

Rademacher Complexity by VC Dimension

Let d_{VC} be the VC dimension of a hypothesis set $\mathcal{H}$. Then for any set $S = \{(\mathbf{x}_1, \ldots, \mathbf{x}_n)\}$ we have

$$\mathrm{R}_S(\mathcal{H}) \lesssim \left(\frac{d_{\mathrm{VC}} \log n}{n}\right)^{\frac{1}{2}}.$$

Proof. Let $A = \left\{(h(\mathbf{x}_1), \ldots, h(\mathbf{x}_n)) : h \in \mathcal{H}\right\}$. By Massart's Lemma, we know

$$\mathfrak{R}_S(\mathcal{H}) = \mathbb{E}\Big[\sup_{h\in\mathcal{H}} \frac{1}{n}\sum_{i=1}^{n} \epsilon_i h(\mathbf{x}_i)\Big] = \mathbb{E}\Big[\sup_{\mathbf{a}\in A} \frac{1}{n}\sum_{i=1}^{n} \epsilon_i a_i\Big] \leq \frac{\left(2n \log m_{\mathcal{H}}(n)\right)^{\frac{1}{2}}}{n}$$

$$\leq \frac{\left(2n \log(n^{d_{\mathrm{VC}}} + 1)\right)^{\frac{1}{2}}}{n} \lesssim \left(\frac{d_{\mathrm{VC}} \log n}{n}\right)^{\frac{1}{2}}.$$

Covering Number

Covering Numbers

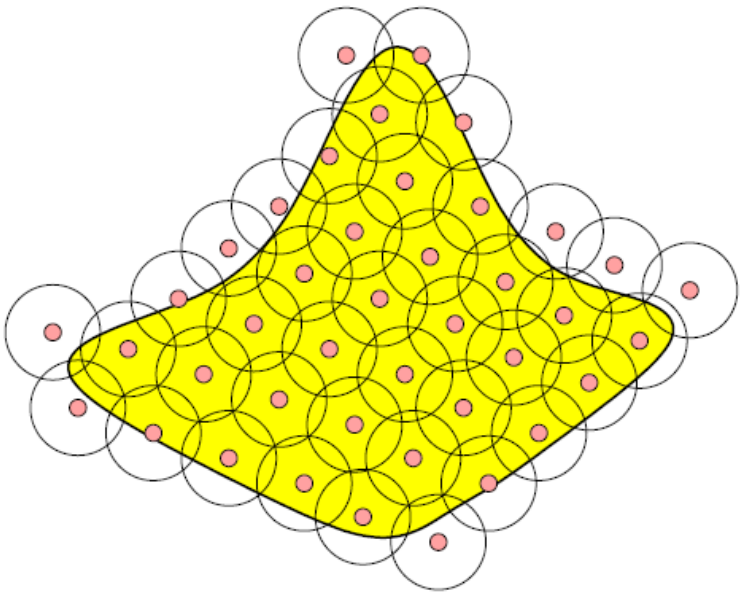
- Consider a class $\mathcal{F}$ of real-valued functions defined over $\mathcal{Z}$
- We **project** $\mathcal{F}$ onto the dataset $S = \{z_1, \ldots, z_n\}$

$$\Big\{\big(f(\mathbf{z}_1), f(\mathbf{z}_2), \ldots, f(\mathbf{z}_n)\big) : f \in \mathcal{F}\Big\}$$

- $\{f^{(1)}, \ldots, f^{(m)}\}$ is an ϵ-cover of $\mathcal{F}$ w.r.t. S and L_p-norm if

$$\sup_{f \in \mathcal{F}} \min_{j \in [m]} \underbrace{\left(\frac{1}{n}\sum_{i=1}^{n} |f(\mathbf{z}_i) - f^{(j)}(\mathbf{z}_i)|^p\right)^{\frac{1}{p}}}_{\text{distance w.r.t. } \|\cdot\|_p} \leq \epsilon \tag{12}$$

- $\mathcal{N}(\epsilon, \mathcal{F}, d_p)$: the **smallest** cardinality of an ϵ-cover for $\mathcal{F}$ w.r.t. S



$$d_p(f, g) = \left(\frac{1}{n}\sum_{i=1}^{n} |f(\mathbf{z}_i) - g(\mathbf{z}_i)|^p\right)^{\frac{1}{p}}$$

- $p = 2$: $d_2(f, g) = \left(\frac{1}{n}\sum_{i=1}^{n} |f(\mathbf{z}_i) - g(\mathbf{z}_i)|^2\right)^{\frac{1}{2}}$
- $p = \infty$: $d_\infty(f, g) = \max_{i \in [n]} |f(\mathbf{z}_i) - g(\mathbf{z}_i)|$

Covering Number

- Covering number measures the capacity of a function space by the number of balls to approximate the function space to a specified accuracy
- We first project the functions to S and get a set of vectors in $\mathbb{R}^n$
- We then measure the capacity by L_p norm in the vector class

There is a close connection between covering number and Rademacher complexity!

Lipschitzness

We say $f : \mathcal{W} \mapsto \mathbb{R}$ is G-Lipschitz continuous if

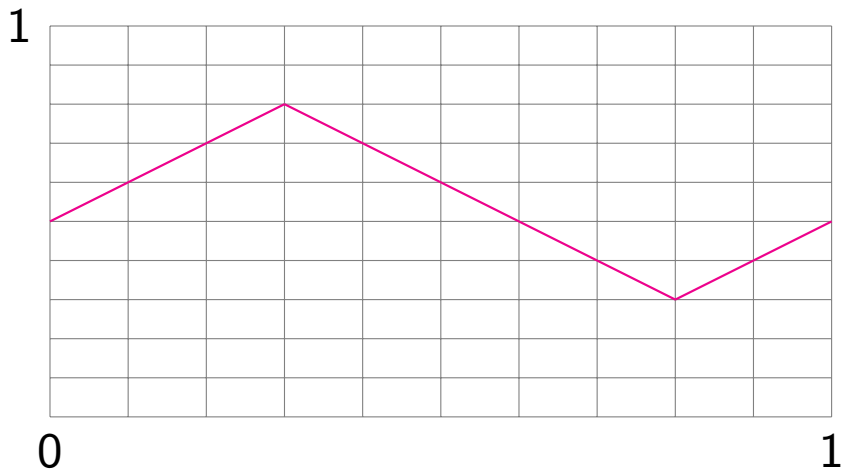
$$|f(\mathbf{w}) - f(\mathbf{w}')| \leq G\|\mathbf{w} - \mathbf{w}'\|_2.$$

Covering Number Estimates: 1-d Lipschitz functions

Let $\mathcal{F}$ be the set of G-Lipschitz functions mapping from $[0,1]$ to $[0,1]$, then

$$\log \mathcal{N}(\epsilon, \mathcal{F}, d_\infty) \lesssim G/\epsilon.$$

- Form an ϵ-grid of the y-axis, and an ϵ/G grid of the x-axis



- Consider all functions that are piecewise linear on this grid, where all pieces have slops $+G$ or $-G$
- There are $1/\epsilon$-starting points, and for each starting point there are $2^{G/\epsilon}$ slop choices
- The set of all such piecewise linear functions form an $O(\epsilon)$-cover.
- The cardinality of this set is $\frac{2^{G/\epsilon}}{\epsilon}$. Therefore,

$$\log \mathcal{N}(\epsilon, \mathcal{F}, d_\infty) \leq \log 2^{G/\epsilon}/\epsilon \lesssim G/\epsilon$$

Covering Number Estimates: Lipschitz functions

d-dimensional Lipschitz functions

Let $\mathcal{F}_d$ be the set of G-Lipschitz functions mapping from $[0,1]^d$ to $[0,1]$, then

$$\log \mathcal{N}(\epsilon, \mathcal{F}, d_\infty) \lesssim G^d/\epsilon^d.$$

Note the exponential dependency on the dimension

Covering Number: Examples

$$\text{Let} \quad A_\infty = \left\{\mathbf{x} \mapsto \mathbf{w}^\top \mathbf{x} : \mathbf{w} \in \mathbb{R}^d, \|\mathbf{w}\|_\infty \le 1\right\}, \text{ assume } \|\mathbf{x}\|_1 \le 1.$$

- We need to find a α-cover of A_∞ under d_∞.
- For any $f, g \in A_\infty$, there exist $\mathbf{w}_f$ and $\mathbf{w}_g$ such that $f(\mathbf{x}) = \mathbf{w}_f^\top \mathbf{x}$ and $g(\mathbf{x}) = \mathbf{w}_g^\top \mathbf{x}$:

$$\max_{i\in[n]} |f(\mathbf{x}_i) - g(\mathbf{x}_i)| = \max_{i\in[n]} |\mathbf{w}_f^\top \mathbf{x}_i - \mathbf{w}_g^\top \mathbf{x}_i| = \max_{i\in[n]} |(\mathbf{w}_f - \mathbf{w}_g)^\top \mathbf{x}_i| \le \|\mathbf{w}_f - \mathbf{w}_g\|_\infty.$$

- It suffices to find a set of vectors $\mathcal{C} \subset \mathbb{R}^d$ s.t. for any $\mathbf{w} \in \mathcal{W} = \{\mathbf{w} \in \mathbb{R}^d : \|\mathbf{w}\|_\infty \le 1\}$ there exists $\tilde{\mathbf{w}}$ with $\|\mathbf{w} - \tilde{\mathbf{w}}\|_\infty \le \alpha$.
- It is clear the following $\mathcal{C}$ covers $\mathcal{W}$ with accuracy α under the metric $d(\mathbf{w}, \mathbf{w}') := \max_i |w^{(i)} - (w')^{(i)}|$:

$$\mathcal{C} = \left\{(w^{(1)}, \ldots, w^{(d)}) : w^{(i)} \in \{0, \alpha, 2\alpha, \ldots, \lceil 1/\alpha \rceil \alpha, -\alpha, -2\alpha, \ldots, -\lceil 1/\alpha \rceil \alpha\}\right\}.$$

- Furthermore, $|\mathcal{C}| \le \left(2\lceil 1/\alpha \rceil + 1\right)^d \le (3/\alpha)^d$

$$\log \mathcal{N}(\alpha, A_\infty, d_\infty) \le d \log(3/\alpha).$$

Bounding Rademacher Complexity by Covering Number

- For both Rademacher complexity and covering number, we project $\mathcal{F}$ to S and get

$$\left\{\left(f(\mathbf{z}_1), f(\mathbf{z}_2), \ldots, f(\mathbf{z}_n)\right) : f \in \mathcal{F}\right\}$$

- The projection onto S matters and we introduce the notation

$$\tilde{f} = \left(f(\mathbf{z}_1), f(\mathbf{z}_2), \ldots, f(\mathbf{z}_n)\right) \in \mathbb{R}^n, \quad f \in \mathcal{F}.$$

$$\Longrightarrow \mathfrak{R}_S(\mathcal{F}) = \frac{1}{n}\mathbb{E}_\epsilon\left[\sup_{f\in\mathcal{F}}\langle \boldsymbol{\epsilon}, \tilde{f}\rangle\right].$$

- We will provide two connections between Rademacher complexities and covering numbers
  - one-step discretization
  - chaining argument

One-step Discretization

One-step discretization. If $|f(x)| \le 1$, then

$$\mathfrak{R}_S(\mathcal{F}) \le \inf_{\alpha} \left(\alpha + \left(\frac{2}{n} \log \mathcal{N}(\alpha, \mathcal{F}, d_2) \right)^{\frac{1}{2}} \right).$$

- Let $\mathcal{F}_\alpha$ be an α-cover of $\mathcal{F}$ w.r.t. L_2-norm.
- Let $\pi(f) \in \mathcal{F}_\alpha$ satisfy $d_2(f, \pi(f)) \le \alpha$. Then, $\|\tilde{f} - \widetilde{\pi(f)}\|_2 \le \sqrt{n}\alpha$

$$\begin{aligned}
\mathfrak{R}_S(\mathcal{F}) &= \frac{1}{n}\mathbb{E}_\epsilon \sup_{f \in \mathcal{F}} \left[\langle \boldsymbol{\epsilon}, \tilde{f} - \widetilde{\pi(f)} \rangle + \langle \boldsymbol{\epsilon}, \widetilde{\pi(f)} \rangle \right] \\
&\le \frac{1}{n}\mathbb{E}_\epsilon \sup_{f \in \mathcal{F}} \left[\langle \boldsymbol{\epsilon}, \tilde{f} - \widetilde{\pi(f)} \rangle \right] + \frac{1}{n}\mathbb{E}_\epsilon \sup_{f \in \mathcal{F}} \left[\langle \boldsymbol{\epsilon}, \widetilde{\pi(f)} \rangle \right] \\
&\le \frac{1}{n}\mathbb{E}_\epsilon \|\boldsymbol{\epsilon}\|_2 \left\| \tilde{f} - \widetilde{\pi(f)} \right\|_2 + \frac{1}{n}\mathbb{E}_\epsilon \sup_{f \in \mathcal{F}_\alpha} \left[\langle \boldsymbol{\epsilon}, \tilde{f} \rangle \right] \\
&\le \alpha \left(n^{-1}\mathbb{E}[\|\epsilon\|_2^2] \right)^{\frac{1}{2}} + \mathfrak{R}_S(\mathcal{F}_\alpha) \le \alpha + \left(\frac{2}{n} \log |\mathcal{F}_\alpha| \right)^{\frac{1}{2}} \quad \text{(Massart's lemma)}
\end{aligned}$$

Chaining Argument (Optional)

- Let $\mathcal{F}_j$ be an α_j-cover of $\mathcal{F}$ with $\alpha_j = 2^{-j} \cdot D$.
- Let $f_j \in \mathcal{F}_j$ satisfy $d_2(f_j, f) \leq \alpha_j \Longrightarrow \|\tilde{f} - \tilde{f}_j\|_2 \leq \sqrt{n}\alpha_j$. Then

$$d_2(f_j, f_{j-1}) \leq d_2(f_j, f) + d_2(f, f_{j-1}) \leq \alpha_j + \alpha_{j-1} = 3\alpha_j \tag{13}$$

- Let $f_0 = 0$ and consider the decomposition

$$f = f - f_m + \sum_{j=1}^{m} (f_j - f_{j-1}). \tag{14}$$

$$n\mathfrak{R}_S(\mathcal{F}) = \mathbb{E} \sup_{f \in \mathcal{F}} \langle \boldsymbol{\epsilon}, \tilde{f} \rangle = \mathbb{E} \sup_{f \in \mathcal{F}} \left[\langle \boldsymbol{\epsilon}, \tilde{f} - \tilde{f}_m \rangle + \sum_{j=1}^{m} \langle \boldsymbol{\epsilon}, \tilde{f}_j - \tilde{f}_{j-1} \rangle \right]$$

$$\leq \mathbb{E}[\|\boldsymbol{\epsilon}\|_2 \sup_{f \in \mathcal{F}} \|\tilde{f} - \tilde{f}_m\|_2] + \mathbb{E} \sup_{f \in \mathcal{F}} \sum_{j=1}^{m} \langle \boldsymbol{\epsilon}, \tilde{f}_j - \tilde{f}_{j-1} \rangle$$

$$\leq n\alpha_m + \sum_{j=1}^{m} \mathbb{E} \sup_{f \in \mathcal{F}} \langle \boldsymbol{\epsilon}, \tilde{f}_j - \tilde{f}_{j-1} \rangle$$

$$n\mathfrak{R}_S(\mathcal{F}) \leq n\alpha_m + \sum_{j=1}^{m} \mathbb{E} \sup_{\mathbf{a} \in A_j} \langle \boldsymbol{\epsilon}, \mathbf{a} \rangle, \text{ where } A_j := \left\{ \tilde{f}_j - \tilde{f}_{j-1} : f \in \mathcal{F} \right\}.$$

Chaining Argument (Optional)

We just derived

$$n\mathfrak{R}_S(\mathcal{F}) \le n\alpha_m + \sum_{j=1}^{m} \mathbb{E} \sup_{\mathbf{a}\in A_j} \langle \boldsymbol{\epsilon}, \mathbf{a} \rangle, \text{ where } A_j := \left\{ \tilde{f}_j - \tilde{f}_{j-1} : f \in \mathcal{F} \right\}$$

$$d_2(f_j, f_{j-1}) \le 3\alpha_j, \quad \forall f \in \mathcal{F}.$$

Then for $\mathbf{a} = \tilde{f}_j - \tilde{f}_{j-1}$ we have

$$\|\mathbf{a}\|_2^2 = \sum_{i=1}^{n} \left(f_j(\mathbf{z}_i) - f_{j-1}(\mathbf{z}_i) \right)^2 \le n d_2^2(f_j, f_{j-1}) \le n(3\alpha_j)^2 \Longrightarrow \|\mathbf{a}\|_2 \le 3\sqrt{n}\alpha_j \quad \forall \mathbf{a} \in A_j.$$

Cardinality : $\left|A_j\right| \le |\mathcal{F}_j| \cdot |\mathcal{F}_{j-1}| \le \mathcal{N}(\alpha_j, \mathcal{F}, d_2)\mathcal{N}(\alpha_{j-1}, \mathcal{F}, d_2) \le \mathcal{N}^2(\alpha_j, \mathcal{F}, d_2)$

Massart lemma : $\mathbb{E} \sup_{\mathbf{a}\in A_j} \langle \boldsymbol{\epsilon}, \mathbf{a} \rangle \le 3\sqrt{n}\alpha_j \left(4 \log \mathcal{N}(\alpha_j, \mathcal{F}, d_2) \right)^{\frac{1}{2}} = 6\alpha_j \sqrt{n} \log^{\frac{1}{2}} \mathcal{N}(\alpha_j, \mathcal{F}, d_2).$

$$\Longrightarrow \mathfrak{R}_S(\mathcal{F}) \le \alpha_m + 6 \sum_{j=1}^{m} \alpha_j \left(\frac{\log \mathcal{N}(\alpha_j, \mathcal{F}, d_2)}{n} \right)^{\frac{1}{2}}.$$

Chaining Argument (Optional)

$$\text{We derived} \quad \mathfrak{R}_S(\mathcal{F}) \le \alpha_m + 6\sum_{j=1}^{m} \alpha_j \left(\frac{\log \mathcal{N}(\alpha_j, \mathcal{F}, d_2)}{n}\right)^{\frac{1}{2}}.$$

Since $\alpha_j - \alpha_{j+1} = \alpha_j/2$, we know

$$\sum_{j=1}^{m} \alpha_j \log^{\frac{1}{2}} \mathcal{N}(\alpha_j, \mathcal{F}, d_2) = 2\sum_{j=1}^{m} (\alpha_j - \alpha_{j+1}) \log^{\frac{1}{2}} \mathcal{N}(\alpha_j, \mathcal{F}, d_2)$$

$$= 2\sum_{j=1}^{m} \int_{\alpha_{j+1}}^{\alpha_j} \log^{\frac{1}{2}} \mathcal{N}(\alpha_j, \mathcal{F}, d_2) \mathrm{d}\alpha = 2\sum_{j=1}^{m} \int_{\alpha_{j+1}}^{\alpha_j} \log^{\frac{1}{2}} \mathcal{N}(\alpha, \mathcal{F}, d_2) \mathrm{d}\alpha$$

$$\le 2\int_{\alpha_{m+1}}^{D} \log^{\frac{1}{2}} \mathcal{N}(\alpha, \mathcal{F}, d_2) \mathrm{d}\alpha.$$

Dudley's chaining integral. If $d_2(f, 0) \le D$ for all f. Then

$$\mathfrak{R}_S(\mathcal{F}) \le \min_x \left[4x + 12\int_x^D \left(\frac{\log \mathcal{N}(\alpha, \mathcal{F}, d_2)}{n}\right)^{\frac{1}{2}} d\alpha\right].$$

Connecting Covering Number to Rademacher Complexity

- If $\mathcal{N}(\alpha, \mathcal{F}, d_2) \lesssim \alpha^{-R}$, then $\log \mathcal{N}(\alpha, \mathcal{F}, d_2) \lesssim R \log 1/\alpha$

$$\int_0^1 \left(\frac{\log \mathcal{N}(\alpha, \mathcal{F}, d_2)}{n}\right)^{\frac{1}{2}} \mathrm{d}\alpha \lesssim \int_0^1 \left(\frac{R \log 1/\alpha}{n}\right)^{\frac{1}{2}} \mathrm{d}\alpha \lesssim \left(\frac{R}{n}\right)^{\frac{1}{2}}$$

- If $\mathcal{N}(\alpha, \mathcal{F}, d_2) \lesssim B^{R/\alpha}$, then $\log \mathcal{N}(\alpha, \mathcal{F}, d_2) \lesssim \frac{1}{\alpha} R \log B$

$$\int_0^1 \left(\frac{\log \mathcal{N}(\alpha, \mathcal{F}, d_2)}{n}\right)^{\frac{1}{2}} \mathrm{d}\alpha \lesssim \int_0^1 \left(\frac{R \log B}{n\alpha}\right)^{\frac{1}{2}} \mathrm{d}\alpha \lesssim \left(\frac{R}{n}\right)^{\frac{1}{2}}$$

- If $\mathcal{N}(\alpha, \mathcal{F}, d_2) \lesssim B^{R/\alpha^2}$, then $\log \mathcal{N}(\alpha, \mathcal{F}, d_2) \lesssim \frac{1}{\alpha^2} R \log B$

$$\int_0^1 \left(\frac{\log \mathcal{N}(\alpha, \mathcal{F}, d_2)}{n}\right)^{\frac{1}{2}} \mathrm{d}\alpha \lesssim \int_0^1 \left(\frac{R \log B}{n\alpha^2}\right)^{\frac{1}{2}} \mathrm{d}\alpha = \left(\frac{R \log B}{n}\right)^{\frac{1}{2}} \int_0^1 \frac{1}{\alpha} \mathrm{d}\alpha = \infty$$

However, $\alpha_m + \int_{\alpha_m}^1 \left(\frac{\log \mathcal{N}(\alpha, \mathcal{F}, d_2)}{n}\right)^{\frac{1}{2}} \mathrm{d}\alpha \lesssim \alpha_m + \left(\frac{R \log B}{n}\right)^{\frac{1}{2}} \int_{\alpha_m}^1 \frac{1}{\alpha} \mathrm{d}\alpha$

$$= \alpha_m + \left(\frac{R \log B}{n}\right)^{\frac{1}{2}} \log \frac{1}{\alpha_m} = \widetilde{O}\left(\left(\frac{R \log B}{n}\right)^{\frac{1}{2}}\right) \left(\alpha = \left(\frac{R \log B}{n}\right)^{\frac{1}{2}}\right)$$

Summary

Summary

- Population risk and empirical risk
- Error decomposition:
  - Optimization error: limited computational power
  - Generalization error: limited data
- Concentration inequality
- Complexity measures of function spaces
  - Rademacher complexity
  - Growth function and VC dimension
  - Covering number